

Neuroprotective effects of *Moringa oleifera* leaf extract via potential NF- κ B modulation in cuprizone-induced demyelination in rats: Integrated *in vivo* and *in silico* evidence

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ABSTRACT

Myelin sheath impairment is linked to several neurological disorders; however, effectively counteracting its pathological consequences remains a formidable challenge. Herein, we investigate the neuroprotective potential of *Moringa oleifera* leaf extract (MOLE) against cuprizone (CPZ)-instigated demyelination in the corpus callosum (CC) of rats. Phytochemical profiling of MOLE was analyzed using gas chromatography-mass spectrometry (GC-MS). For *in vivo* investigations, adult male *Wistar* albino rats were assigned into four groups: Group I (control) received corn oil 0.5 mL and distilled water (1 mL) daily by oral gavage for three weeks; Group II (MOLE) received MOLE (300 mg/kg/day) in 1 mL distilled water orally for three weeks; Group III (CPZ) received CPZ (600 mg/kg/day) in 0.5 mL corn oil for one week, followed by 200 mg/kg/day for the next two weeks; Group IV (CPZ + MOLE) received both CPZ and MOLE according to the regimens described for Groups II and III. Neurobehavioral assessments using the rotarod and inverted screen tests revealed notable aberrations in CPZ-treated rats compared to control rats, whereas MOLE administration markedly ameliorated these deficits. Biochemically, CPZ treatment markedly deregulated oxidative stress markers in the CC, increasing malondialdehyde (MDA) and nitric oxide (NO), while reducing reduced glutathione (GSH), superoxide dismutase (SOD), and catalase (CAT) activities. Consistently, it elevated inflammatory biomarkers, including tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), interleukin-6 (IL-6), and nuclear factor kappa B (NF- κ B), accompanied by histological and ultrastructural anomalies compared with controls and those treated only with MOLE, suggesting its safety profile. Interestingly, co-administration of MOLE normalized oxidative stress and inflammatory markers, potentially through NF- κ B orchestration, and correlated with histopathological and ultrastructural improvements. Molecular docking investigations further showed robust binding of bioactive compounds in MOLE within TNF- α and NF- κ B active sites, substantiating *in vivo* findings. Collectively, our findings accentuate the potential prophylactic potency of MOLE against CPZ-induced demyelination.

1. Introduction

Demyelinating disorders are a diverse category of neurological diseases defined by progressive damage or degeneration of the myelin sheath within the central or peripheral nervous system (Bernardo and

Visentin, 2023). These disorders impair nerve impulse transmission and contribute not only to physical disability but also to neurological disorders and reduced quality of life (Murayama et al. 2025). Primary demyelination is classically represented by multiple sclerosis (MS) as the most common disorder, whereas secondary demyelination may

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Table 1
Operation conditions of GC-MS.

Operation Parameter	Value
Carrier gas	He _(gas) (average velocity 39 cms ⁻¹), 1 mLmin ⁻¹ flow rate.
Oven temperature program	60 °C (1 min), 110 °C (2 min, 9 °C min ⁻¹), 110–240 °C (5 min., 7 °C min ⁻¹), and 240–260 °C (1 min, 20 °C min ⁻¹).
Column type	Thermo TG-5MS Zebron capillary coiled column (length 30 m x 0.25 mm ID, 0.25 µm film thickness).
Injection type& ratio	Split, 10:1.
EI, scan range	EI 70 eV, with a scanning range from 50 to 600 m/z (five scans/sec).

engender a broad spectrum of deregulations, including ischemic, infectious, metabolic, and other neurological conditions (Barkhof and Koeller, 2020, Jiang et al. 2025). Common manifestations of MS include visual impairment, motor dysfunction, sensory abnormalities and tingling, and cognitive decline (Kuhlmann et al. 2023).

It is recognized that oligodendrocytes (OLs) are responsible for axon myelination in the brain and spinal cord (Abbas et al. 2021). In MS, myelin produced by oligodendrocytes in the central nervous system, as well as by Schwann cells in the peripheral nervous system, undergoes degeneration as a result of inflammation, glial cell damage, metabolic disturbances, and viral or immune-mediated impairments (Lubetzki and Stankoff, 2014, Höftberger and Lassmann, 2018, Coutinho Costa et al. 2023, Zhao and Jacob, 2023). These are associated with mitochondrial dysfunction, impaired myelin repair, and axonal degeneration (Lassmann, 2018, Delic et al. 2025). Increasing evidence indicates that oxidative stress and inflammation are central contributors to demyelinating pathology, as oxidative injury damages proteins, lipids, and

DNA in neurons and glial cells, thereby provoking cellular dysfunction and tissue degeneration (Tobore, 2021). In parallel, mitochondrial dysfunction further disrupts redox homeostasis, energy metabolism, and neuronal integrity, thereby aggravating disease progression (Kim et al. 2015, Houldsworth, 2024).

The burden of demyelinating diseases is substantial globally. In North Africa, MS prevalence in particular has increased considerably, while Egypt reports the largest number of cases (about 60,000), with recent estimates indicating a prevalence of 25 cases/100,000 (Zakaria et al. 2020, Rivera, 2026). Furthermore, a recent epidemiological study highlighted the escalating prevalence of MS across several countries, reporting the highest percentage increases in Algeria (118%), Taiwan (300%), Qatar (118%), Egypt (158%), Norway (80%), France (62%), and Spain (84%) (Khan and Hashim, 2025). In North America, Canada and the United States are also among the regions with the highest MS prevalence worldwide, reporting 1 million people with MS in the U.S. (Wallin et al. 2019, Rivera, 2026). Consequently, recent therapeutic approaches have been applied to treat demyelinating disorders, including biomodulators, immunosuppressive agents, monoclonal antibodies, remyelination-inducing drugs, and several traditional Chinese medicines (Jiang et al. 2025). Nevertheless, these therapies are largely non-specific, may compromise immune function, increase susceptibility to opportunistic infections, and predominantly focus on disease management rather than cure. For instance, administration of Ublituximab over five years sustained clinical benefit in patients with MS; however, its therapeutic role remains centered on disease management (Cree et al. 2026).

This underscores the imperative to identify novel, safe, naturally derived therapies that mitigate MS-associated neuroinflammation and neurodegeneration, preserve myelin integrity, and promote

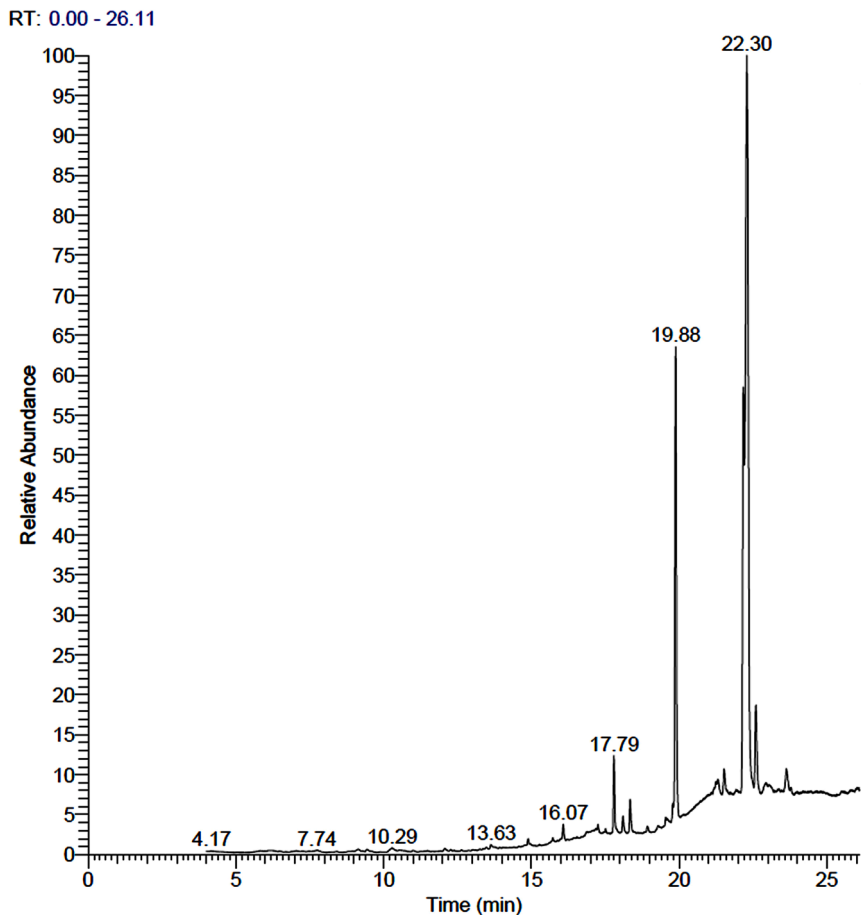


Fig. 1. GC-MS spectrum of ethanolic MOLE. Sharp peaks indicate the purity and stability of extracted phytochemicals. NL 6.12E8, TIC MS3, Rt (min.) 0.00–26.11.

Table 2
Phytochemical compounds of MOLE identified by GC-MS analysis.

Rt	Probability	Compound Name	Area %	Molecular Formula	Molecular Weight
17.24	17.10	Ethyl 13-CH ₃ -tetradecanoate	1.51	C ₁₇ H ₃₄ O ₂	270
	8.30	CH ₃ -2,8-di- CH ₃ -tridecanoate		C ₁₆ H ₃₂ O ₂	256
	7.01	n-Hexadecanoic acid			
17.80	5.55	9-Undecenol, 2,10-di- CH ₃	3.76	C ₁₃ H ₂₆ O	198
	5.12	Ethanol, 2-(9-octadecenyl)-, (Z)-		C ₂₀ H ₄₀ O ₂	312
	3.82	CH ₃ -tetrahydroionol		C ₁₄ H ₂₈ O	212
18.09	15.60	3,7,11,15-tetra- CH ₃ -2-hexadecen-1-ol	1.32	C ₂₀ H ₄₀ O	296
	4.58	5-nonadecen-1-ol		C ₁₉ H ₃₈ O	282
	3.42	Z-7-Octadecen-1-ol acetate		C ₂₀ H ₃₈ O ₂	310
18.34	30.31	Phytol, acetate	2.72	C ₂₂ H ₄₂ O ₂	338
	19.58	3,7,11,15-tetra- CH ₃ -2-hexadecen-1-ol		C ₂₀ H ₄₀ O	296
	8.91	Phytol		C ₂₀ H ₄₀ O	
19.57	8.21	CH ₃ -5,11,14,17-eicosatetraenoate	0.96	C ₂₁ H ₃₄ O ₂	318
	4.73	7,10,13-eicosatrienoic acid, CH ₃ ester		C ₂₁ H ₃₆ O ₂	320
	3.34	Ethyl 5,8,11,14,17-icosapentaenoate		C ₂₂ H ₃₄ O ₂	330
19.91	21.97	Hexadecenoic acid, ethyl ester	17.16	C ₁₈ H ₃₆ O ₂	284
	16.39	Octadecanoic acid			
	20.38	Corynan-17-ol, 18,19-didehydro-10-OCH ₃ -acetate		C ₂₂ H ₂₈ N ₂	368
21.18	19.59	Curan-17-oic acid, 19-(acetyloxy)-2,16-didehydro-20-OH, CH ₃ -ester (19S)-	0.92	C ₂₂ H ₂₆ N ₂ O ₅	398
	5.87	Corynan-17-ol, 18,19-didehydro-10-methoxy-	0.92	C ₂₀ H ₂₆ N ₂ O ₂	326
	9.50	Phytol	2.63	C ₂₀ H ₄₀ O	296
22.20	6.32	E-8-CH ₃ -9-tetradecen-1-ol acetate	2.63	C ₁₇ H ₃₂ O ₂	268
	10.77	Butyl 9,12-octadecadienoate	11.39	C ₂₂ H ₄₀ O ₂	336
	8.25	n-propyl 9,12-octadecadienoate	11.39	C ₂₁ H ₃₈ O ₂	322
22.38	7.93	CH ₃ -7,12-octadecadienoate	11.39	C ₁₉ H ₃₄ O ₂	294
	8.12	n-Propyl 9,12,15-octadecatrienoate	51.23	C ₂₁ H ₃₆ O ₂	320
	7.18	9,12,15-octadecatrienoic acid, ethyl ester, (Z, Z, Z)-	51.23	C ₂₀ H ₃₄ O ₂	306
22.51	7.18	Butyl 9,12,15-octadecatrienoate	51.23	C ₂₂ H ₃₈ O ₂	334
	25.41	9-octadecen-12-ynoic acid, CH ₃ ester	1.71	C ₁₉ H ₃₂ O ₂	292
	8.51	6,9,12-octadecatrienoic acid, CH ₃ ester	1.71		
22.63	5.67	CH ₃ -5,11,14-eicosatrienoate	1.71	C ₂₁ H ₃₆ O ₂	320
	12.97	Eicosanoid acid	3.46	C ₂₀ H ₄₀ O ₂	312
	7.86	Octadecanoic acid, 17- CH ₃ -, CH ₃ ester	3.46		
24.84	6.33	Naphthalene-2,6-di-COOH, pentyl ester 4-pentyl-Ph-ester	3.46	C ₂₈ H ₃₂ O ₄	432
	30.27	1H-cyclopropa [3,4] benz[1,2-e] azulene-4a,5,7b,9,9a(1aH)-pentol 3-[(acetyloxy) CH ₃ -1b,4,5,7a,8,9-hexa-H-1,1,6,8-tetra-CH ₃ , 5,9,9a-triacetate, 1aR-1aà,1bà,4aà,5à,7aà,7bà,8à,9à, 9aà]]	0.62	C ₂₈ H ₃₈ O ₁₀	534
	27.92	1H-Cyclopropa [3,4] benz[1,2-e] azulene-5,7b,9,9a-tetrol, 3-[(acetyloxy)CH ₃]-1a,1b,4,4a,5,7a,8,9-octa-H-1,1,6,8-tetr-CH ₃ , 5,9,9a-triacetate, [1aR-1aà,1bà,4aà,5à, 7aà,7bà,8à,9à,9aà]]	0.62	C ₂₈ H ₃₈ O ₉	518
24.91	5.84	2,3,3',4'-tetra-OCH ₃ -5-(3-OCH ₃ prop-1-enylstilbene	0.62	C ₂₂ H ₂₆ O ₅	370
	19.48	Fenvalerate	0.26	C ₂₅ H ₂₂ ClN O ₃	419
	17.21	L-Serinamide		C ₃₄ H ₄₈ N ₈ O ₆	664
24.97	3.99	1-CH ₃ -5-oxo-L-prolyl-N,1-dimet hyl-L-histidyl-N,1-L-tryptophyl-N, N, N2, O-tetr-CH ₃ -2-heptanone- 6-(2-furanyl)-6-CH ₃ -		C ₁₂ H ₁₈ O ₂	194
	12.06	Azafrin	0.35	C ₂₇ H ₃₈ O ₄	426
	8.51	4H-cyclopropa [5',6'] benz [1',2':7,8] azuleno [5,6-b] oxiren-4-one, 8-(acetyloxy)-1,1a,1b,1c,2a,3,3a,6a, 6b,7,8,8a-dodeca-H-3a,6b,8a-tri-OH-2a-(OH-CH ₃)-1,1,5,7-tetra- CH ₃ -(1aà,1bà,1cà,2aà,3aà,6aà,6bà,7aà,8aà, 8aà)-	0.35	C ₂₂ H ₃₀ O ₈	422
7.85		4H-Cyclopropa [5',6'] benz [1',2':7,8] azuleno [5,6-b] oxiren-4-one, 8,8a-bis(acetyloxy)-2a-[(acetyloxy) CH ₃]-1,1a,1b,1c,2a,3,3a,6a,6b,7,8,8a-dodeca-H-6b-OH-3a-OCH ₃ -1,1,5,7-tetra- CH ₃ -, [1aR-(1aà,1bà,1cà,2aà,3aà,6aà,6bà, 7aà,8aà,8aà)]	0.35	C ₂₇ H ₃₆ O ₁₀	520

remyelination while minimizing adverse effects. As a result, medicinal plants have emerged as promising sources of neuroprotective therapies. Among them, *Moringa oleifera* (*M. oleifera*) is widely recognized for its rich nutraceutical composition, involving proteins, lipids, flavonoids, phenolic acids, and isothiocyanates (Dzuvoor et al. 2022, Chiş et al. 2024). It exhibited various pharmacological, immunomodulatory, and neuroprotective properties (Worku and Tolossa, 2024, Essawy et al. 2026). Additionally, it possesses diverse bioactivities, such as antimicrobial, anti-inflammatory, cardioprotective, antidiabetic, antidepressant, antitumor, antiepileptic, antihypertensive, and antioxidant properties (Klimek-Szczykutowicz et al. 2024, Worku and Tolossa, 2024, Wu et al. 2024, Deepali et al. 2025).

Experimentally, cuprizone (CPZ) administration is broadly utilized to induce demyelinating pathology resembling human myelin disorders (Rokach et al. 2026). As a copper chelator, CPZ disrupts mitochondrial function, amplifying reactive oxygen species (ROS) production, and incites neuroinflammatory responses (Kim et al. 2019). In addition, CPZ damages oligodendrocytes and stimulates astrocytes and microglia,

contributing to innate immune-mediated toxicity (Zirngibl et al. 2022). The CPZ model, therefore, represents a valuable experimental system for investigating demyelination mechanisms and testing potential neuroprotective therapies.

Considering all the above, the current research sought, for the first time, to assess the neuroprotective effects of the Egyptian *M. oleifera* leaf extract (MOLE) against CPZ-induced demyelination in the corpus callosum (CC) region of rats. To achieve this objective, we conducted a comprehensive assessment encompassing neurobehavioral performance, oxidative/antioxidative balance, neuroinflammation, and histological and ultrastructural features. Furthermore, the potential interactions between the major bioactive compounds of MOLE and both NF-κB and TNF-α were evaluated employing molecular docking investigations.

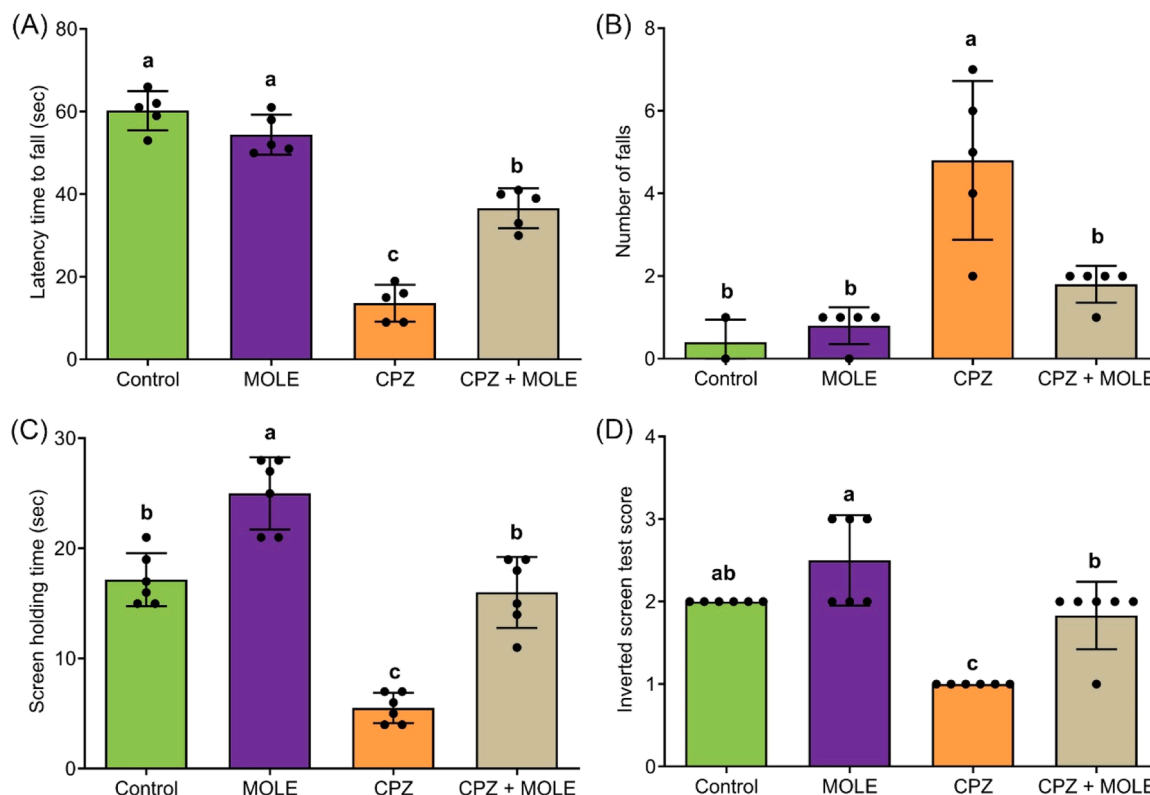


Fig. 2. Behavioral changes as assessed by the rotarod test and the screen holding test in the different experimental groups. (A) latency time to fall in seconds, (B) number of falls, (C) screen holding time in seconds, and (D) inverted screen score. Data are reported as mean $\pm$ SEM ($n = 5-6$). Different letters above the bars ($p < 0.05$) signify statistically significant differences among the animal groups, whereas bars with similar letters (a-c) indicate no statistically significant differences. Abbreviations: CPZ, cuprizone; MOLE, *M. oleifera* leaf extract.

2. Materials and methods

2.1. Chemicals and research kits

Cuprizone (bis-cyclohexanone-oxalyldihydrazone) was acquired from Sigma-Aldrich Co. (Cat. No. C9012-25G). It was dissolved in 0.5 mL of corn oil. Kits for the determination of redox status were obtained from Bio-diagnostic Co. (Egypt). ELISA kits for the measurement of inflammatory markers were obtained from the Bioassay Technology Laboratory (BA Lab, China).

2.2. Plant collection and preparation of *M. oleifera* leaf extract (MOLE)

Leaves of *M. oleifera* were collected from mature trees (aged ca. 10 years) located at the Experimental Station, Faculty of Agriculture, Alexandria University (latitude of 31.205753 and longitude of 29.924525). The plant identification was performed by Prof. Dr. Ahmed A. El-Settawy, professor at the Forestry and Wood Technology Department, Faculty of Agriculture, Alexandria University, Alexandria, Egypt. A weight of 10 g of well-ground powder leaves of *M. oleifera* was dissolved in 100 mL of absolute ethanol and heated under reflux. The mixture was then cooled overnight at room temperature and filtered twice. The residue was dehydrated until it reached a constant weight. The organic layer was dehydrated using anhydrous $MgSO_4$, followed by the evaporation of ethanol from the extraction product by means of a Buchi rotary evaporator under reduced pressure (Mosaad et al. 2025).

2.3. Characterization of MOLE using gas chromatography-mass spectrometry (GC-MS)

Organic chemical compounds in MOLE were identified by means of a

GC-MS (Trace 1300 GC Ultra/Mass Spectrophotometer ISQ QD, Thermo Scientific) equipped with Thermo X-Calibur 2.2 software under operation conditions listed in Table 1. A sample volume (3 μ L) in pure hexane was injected at 275 $^{\circ}$ C into the column and entered the electron impact (EI) ionization chamber for EI by filament. The ions are transferred through ion optics to a quadrupole mass analyzer. The run time was 51.41 min. The GC-MS was integrated with various online mass spectra (MS libraries and NIST databases for several reference compounds) for accurate identification and matching of chemical structures and spectra for the detected separated bioactive components.

2.4. Animal ethical approval

The *in vivo* experimental procedures in this study followed ARRIVE guidelines (<https://arriveguidelines.org>) and the National Institutes of Health guidelines for the care and use of laboratory animals. The Institutional Animal Care and Use Committee (IACUC) at Alexandria University, a member of the International Council for Laboratory Animal Science (ICLAS), approved the investigation protocol (approval number: AU04230328103).

2.5. Animal housing and experimental design

Twenty-four adult male albino Wistar rats (170–190 g; aged 8 weeks) were allocated to the experimental design. The animals were supplied by the animal facility unit of the Faculty of Medicine, Alexandria University, Egypt, housed in stainless steel cages under controlled environmental conditions (temperature set to 23 ± 2 $^{\circ}$ C and 12 h light/dark cycle) with ad libitum access to pelleted diet and tap water. Following one-week acclimatization period, the rats were randomly assigned to four groups (6 rats/group) as follows:

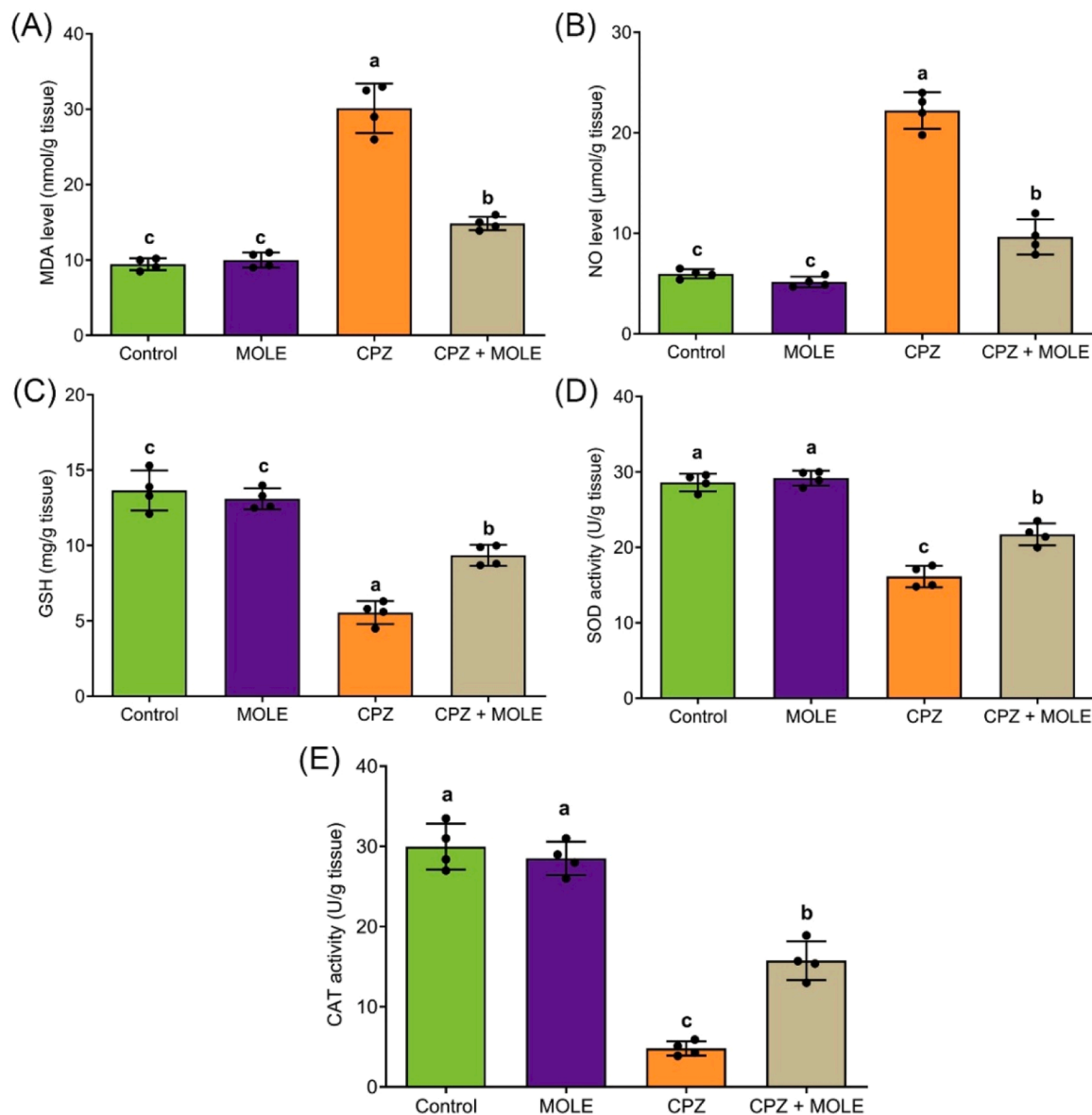


Fig. 3. Assessment of oxidant and antioxidant markers in the CC tissue of rats treated only with CPZ or in combination with MOLE, including (A) MDA, (B) NO, (C) GSH, (D) SOD, and (E) CAT. Data are reported as mean $\pm$ SEM ($n = 4$). Different letters above the bars ($p < 0.05$) signify statistically significant differences among the animal groups, while bars with similar letters (a-c) indicate no statistically significant differences. Abbreviations: CAT, catalase; CPZ, cuprizone; GSH, reduced glutathione; MDA, malondialdehyde; MOLE, *M. oleifera* leaf extract; NO, nitric oxide; SOD, superoxide dismutase.

- Group I (control group) received 0.5 mL corn oil and 1 mL distilled H₂O by oral gavage daily for 3 weeks.
- Group II (MOLE group) received 300 mg/kg/day of MOLE in 1 mL of distilled water via oral gavage daily for 3 weeks, as determined in a previous study (Soliman et al. 2020).
- Group III (CPZ group) was given 600 mg/kg/day of CPZ dissolved in 0.5 mL of corn oil by oral gavage for 1 week, followed by 200 mg/kg/day for the remaining 2 weeks of the experiment. The oral gavage administration protocol for CPZ was applied following previously published studies demonstrating its efficacy in inducing demyelination (Ganji et al. 2020, Essawy et al. 2025) to ensure accurate and consistent dosing among rats and to avoid variability in CPZ intake associated with dietary administration.
- Group IV (CPZ + MOLE) received both CPZ and MOLE as described in the second and third groups.

At the end of the experimental duration, rats underwent neurobehavioral, biochemical, histological, and ultrastructural investigations.

2.6. Neurobehavioral tests

Following the experimental period of 3 weeks, motor coordination and gripping strength of the rat's forelimb and hind limb were assessed using the rotarod and inverted screen behavioral tests, respectively. To minimize animal stress and anxiety, neurobehavioral tests were performed after a 24-hour rest period following completion of the *in vivo* treatments. Before commencing the tests, the rats were acclimated to the behavioral apparatus and trained following the procedures described below. Behavioral test instruments were sanitized with 70% ethanol solution after each trial.

2.6.1. Rotarod test

To assess the effect of CPZ and MOLE on rats' motor balance and movement coordination, the rotarod test was used. In this test, rats were placed on a rotating rod at a height that discouraged jumping off. The primary outcome measures are the rats' latency to fall and the number of falls. Before testing, rats were habituated to the rotarod apparatus by

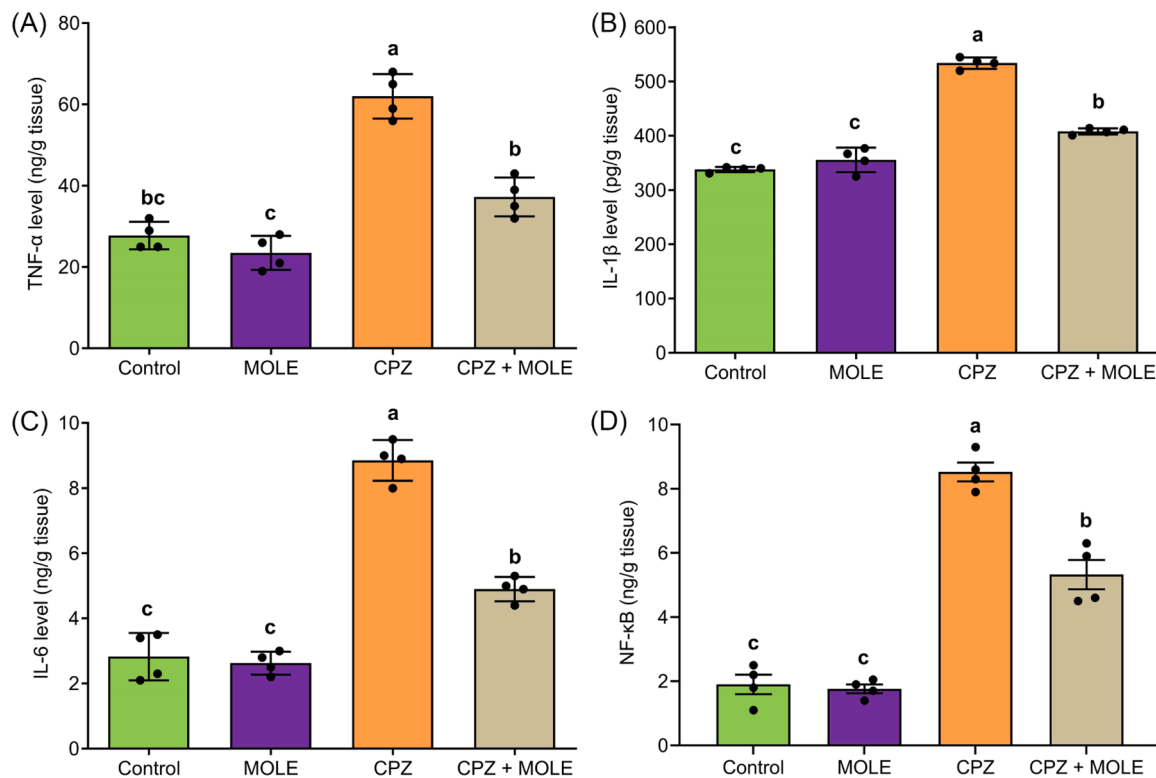


Fig. 4. Changes in the inflammatory markers in the CC tissues of rats treated only with CPZ or in combination with MOLE, including (A) TNF- α , (B) IL-1 β , (C) IL-6, and (D) NF- κ B. Data are reported as mean $\pm$ SEM (n = 4). Different letters above the bars (p < 0.05) signify statistically significant differences among the animal groups, whereas bars with similar letters (a-c) indicate no statistically significant differences. Abbreviations: CPZ, cuprizone; MOLE, *M. oleifera* leaf extract; TNF- α , tumor necrosis factor- α ; IL-1 β , interleukin 1 β ; IL-6, interleukin 6; NF- κ B, nuclear factor kappa B.

training them for 3 days to maintain balance while walking against the rotating drum. During training, the rotational speed increased from 10 to 20 rpm for a maximum test duration of 2 min. Following completion of the training period, each rat was placed individually on the rotating rod, which was positioned at a sufficient height to discourage voluntary jumping. After three days of training, motor coordination was evaluated at a constant rotational speed of 20 rpm, and the latency time to drop off the rotating rod and the number of falls were recorded. Rats that remained on the rod for the entire 300 s test period without falling were removed from the apparatus and were assigned the maximum latency value of 300 s (Pritchett and Mulder, 2003).

2.6.2. Inverted screen test

On the day following the previous test, the inverted screen test was performed to assess the gripping strength of the rat's limb as previously delineated (Essawy Essawy et al. 2024). The screen measured 43 cm on each side and was made from 12 mm wire mesh constructed from 1 mm-diameter wire. To prevent rats from climbing to the opposite side, the mesh-wired screen was bordered by a 4 cm-deep wood border. After a training session for 20 min, each rat was positioned in the center of a wire mesh screen, which was then turned upside down for 2 s, with the rat's head facing downward. The screen was consistently positioned approximately 40–50 cm horizontally on the padding surface. The time was recorded when the rat fell off, or it was removed when the 60 s limit was reached.

2.7. Collection and preparation of samples

Following completion of the neurobehavioral assessments, the animals were fasted overnight. Four rats from each group were randomly selected and euthanized by decapitation after being anesthetized with ketamine (90 mg/kg) and xylazine (10 mg/kg) via intraperitoneal

injection. The brain was promptly excised, followed by a careful dissection of the CC, dividing into two halves, rinsing with cold saline, and storing at -80°C for biochemical studies. One-half of the CC was homogenized on ice in 0.05 M phosphate buffer (pH 7.4) at 10% (w/v) before clarification at 12,000 rpm for 10 min. The supernatants were preserved at -20°C for subsequent evaluation of oxidative stress and inflammatory biomarkers.

For histopathological and ultrastructural analyses, the remaining two rats from each group were transcardially perfused with a 4% PFA prepared in 0.1 M phosphate-buffered saline. The brains were then carefully removed, and the CC was instantly immersed in 4% formaldehyde: 1% glutaraldehyde (4 F:1 G, pH 7.2) fixative buffer for 3 h for subsequent semithin and transmission electron microscopy.

2.8. Evaluation of oxidative and antioxidative markers

The levels of oxidative/antioxidative markers were quantified in the CC tissue homogenates using commercially available assay kits (Bio-diagnostic Co., Giza, Egypt) in accordance with the manufacturer's protocols. Malondialdehyde (MDA, Cat. No. MD2529), as an indicator of lipid peroxidation, was assayed based on the reaction of thiobarbituric acid-reactive substances (TBARS), with absorbance measured at 530 nm following the method of Satoh (1978). Nitric oxide (NO, Cat. No. NO2533) was estimated indirectly by quantifying nitrite, a stable NO metabolite, utilizing the Griess reaction at 540 nm as previously described (Montgomery and Dymock, 1962). Reduced glutathione (GSH, Cat. No. GR2511) was evaluated using Ellman's reagent (5, 5'-dithiobis-(2-nitrobenzoic acid), DTNB), and the resulting yellow chromogen was measured at 405 nm according to Beutler and Kelly (1963). Catalase (CAT, Cat. No. CA2517) activity was determined by measuring the decomposition rate of hydrogen peroxide as previously reported (Aebi, 1984). Superoxide dismutase (SOD, Cat. No. SD2521)

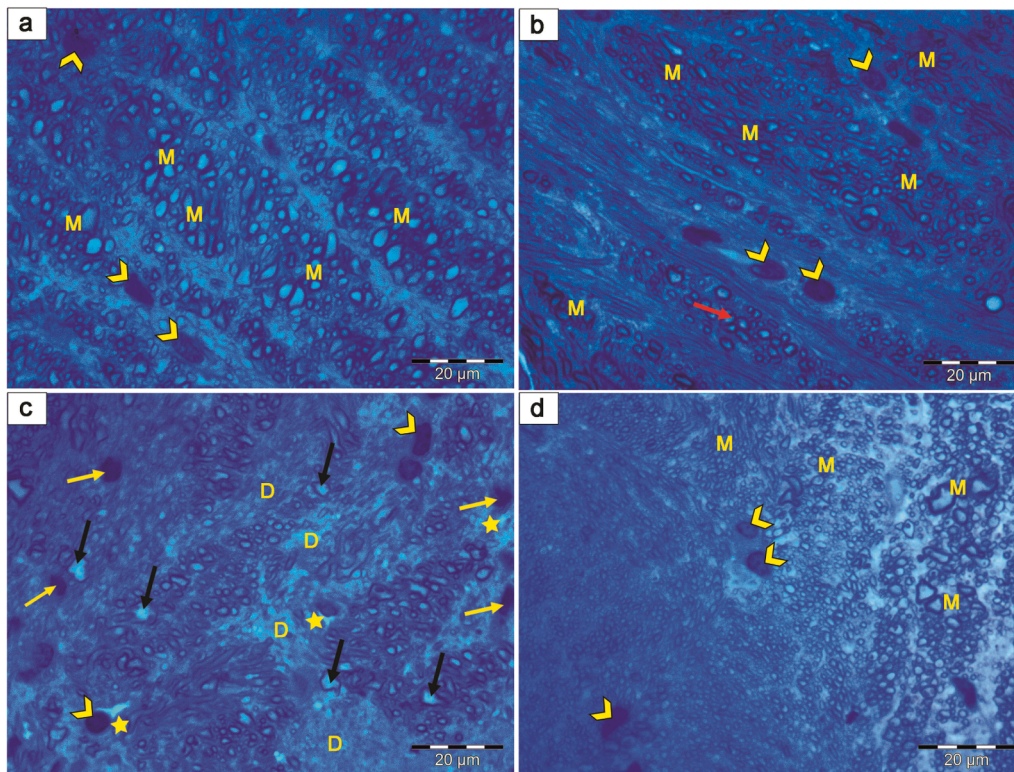


Fig. 5. Photomicrographs illustrate semithin sections of the CC tissue from the brains of rats in different animal groups stained with toluidine blue. **Figs. (5a and 5b)** The control and MOLE-treated groups showed a high number of myelinated nerve fibers (M) and abundant glial cells with normal morphological appearance (arrowheads). **Fig. (5c)** A section from the CPZ-treated group depicted large areas of demyelinated nerve fibers (D), degenerated axons (black arrows), pyknotic glial cells (yellow arrows), gaps (asterisks), and abnormal morphological features of glial cells (arrowheads). **Fig. (5d)** Section of CPZ plus MOLE-treated group revealed remyelination of the majority of nerve fibers (M), and restoration of normal glial cell morphology (arrowheads). Scale bars = 20 µm.

activity was measured based on its capacity to inhibit the reduction of nitroblue tetrazolium (NBT) by superoxide radicals generated in the reaction mixture, following the method of Nishikimi et al. (1972).

2.9. Assessment of neuroinflammatory parameters

Levels of tumor necrosis factor- α (TNF- α , Cat. No. E0764Ra), interleukin-1 β (IL-1 β , Cat. No. E0119Ra), and interleukin-6 (IL-6, Cat. No. E0135Ra) were quantified in the CC tissue homogenates using ELISA kits (BT Lab Co., China) according to the procedures provided by the manufacturer, whereas nuclear factor kappa B (NF- κ B) levels were estimated utilizing the corresponding ELISA kit (Cat. No. ER1186, Fine Test Co., China).

2.10. Semithin and ultrastructure investigations of corpus callosum (CC)

The CC tissues previously fixed in 4 F:1 G fixative were post-fixed in 2% osmium tetroxide for 3 h at 4 °C for semithin and transmission electron microscopy (TEM) processing. For semithin section analysis, the specimens were dehydrated through a graded ethanol series and embedded in an Epon-Araldite resin mixture, followed by sectioning at a thickness of 0.5 µm employing an ultramicrotome (LKB Bromma 2088 Ultratome, Leica Instruments, Bannockburn, IL, USA). Following this, the sections were stained with 1% toluidine blue prepared in 1% sodium tetraborate and then probed with a light microscope (Olympus CX31, Tokyo, Japan).

To investigate the ultrastructural features of the CC tissue, ultrathin sections (60 nm) were mounted on 200-mesh copper grids, visualized with uranyl acetate and lead citrate, and then examined by TEM (JEM-1400 Plus, Tokyo, Japan) operated at an accelerating voltage of 80 kV. We also quantitatively assessed myelin thickness and axon diameter in

ultrastructural sections of CC tissues across different experimental groups via ImageJ Fiji software. Additionally, the G-ratio, which is the ratio of the inner axon diameter to the total outer diameter of the myelinated fiber, was determined (Pegoretti et al. 2025).

2.11. Molecular docking simulation

Molecular docking was undertaken to appraise the potential interactions between the major bioactive constituents of MOLE, hexadecenoic acid ethyl ester, butyl-9,12-octadecadienoate, n-propyl-9,12-octadecadienoate, and methyl-7,12-octadecadienoate, and a key inflammatory mediator, the central transcriptional regulator NF- κ B, and TNF- α . The 2D structures of ligands were compiled from the PubChem database (<https://pubchem.ncbi.nlm.nih.gov>, accessed on 29.10.2025), and converted into 3D conformations, geometry-optimized, and energy-minimized employing Avogadro program version 1.2.0 with the MMFF94 force field (performed on 29.10.2025) (Hanwell et al. 2012). The optimized ligands were prepared in AutoDock Tools 1.5.7 (performed on 29 October 2025) by assigning Gasteiger charges, merging non-polar hydrogens, and defining rotatable bonds. The crystal structures of NF- κ B p50/p65 (PDB ID: 1VKX) and TNF- α (PDB ID: 2AZ5) were sourced from the Protein Data Bank (<https://www.rcsb.org>), and missing residues were modeled using MODELLER within UCSF ChimeraX 1.9 (performed on 30.10.2025) (Webb and Sali, 2017; Meng et al. 2023). NF- κ B p50/p65 and TNF- α were prepared for docking by means of AutoDock Tools version 1.5.7 (performed on 30.10.2025) (Goodsell and Olson, 1990; Morris et al. 2009). Protein preparation included elimination of water molecules, addition of polar hydrogens, and assignment of Kollman charges. Docking was conducted using AutoDock Vina 1.2.3 (performed on 30.10.2025) (Eberhardt et al. 2021).

For NF- κ B p50/p65, the docking grid box was centered on the DNA-

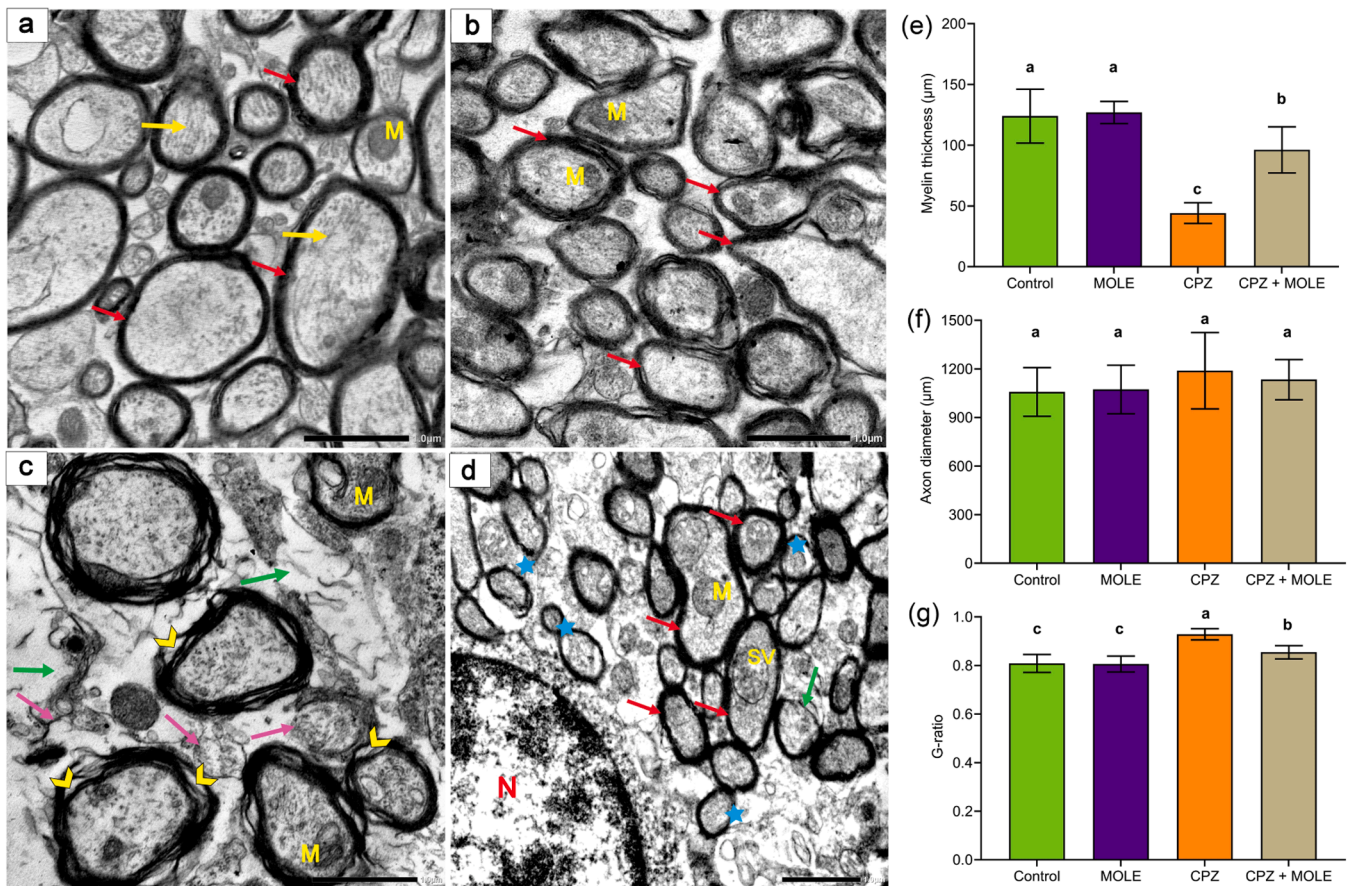


Fig. 6. Transmission electron micrographs (TEM) of the CC tissue from the brains of rats in different animal groups depict the ultrastructural features of myelination and demyelination, characterized by axonal and myelin sheath characteristics. Figs. (6a and 6b) display myelinated nerve fibers (red arrows) in the control and MOLE group, respectively. Normal ultrastructural characteristics of nerve fibers were observed, with darkly stained, compact, and parallel myelin lamellae, axoplasm containing neurofilaments (yellow arrows), and normal mitochondria (M). Fig. (6c) portrays several pathological changes appeared in the CPZ-treated group, characterized by complete and partial axonal demyelination (pink arrows), lamellae displacement marked by the presence of gaps (arrowheads), and axonal damage (green arrows). The mitochondria appear vacuolated with indistinct cristae (M). Fig. (6d) Co-treatment with CPZ and MOLE led to remyelination of nerve fibers, which was observed as thinly myelinated fibers (asterisks) and other tight and compact myelin (red arrows), containing synaptic vesicles (SV) and normal mitochondria (M). A part of an oligodendrocyte's nucleus is shown (N). Scale bars = 1 μm. Fig. (6e-g) Quantitative analysis of myelin thickness, axon diameter, and G-ratio, respectively, in the CC ultrastructural sections following examination of TEM images from different experimental groups, and values are shown as mean ± SD. Different letters above the bars ($p < 0.05$) denote statistically significant differences among groups, while similar letters (a-c) point to no statistically significant difference. Abbreviations: CPZ, cuprizone; MOLE, *M. oleifera* leaf extract.

binding interface defined from the NF-κB-DNA complex. For TNF-α, the grid box was defined based on the coordinates of its co-crystallized ligand (SPD304). Binding interactions were visualized and analyzed with the help of BIOVIA Discovery Studio Visualizer 2024 (performed on 30.10.2025) (Nabil-Adam et al. 2023).

2.12. *In silico* pharmacokinetic investigation

The *in silico* pharmacokinetic properties, including absorption, distribution, metabolism, excretion, toxicity (ADMET), and drug-likeness of the four bioactive compounds identified in MOLE were predicted by means of the pkCSM (<https://biosig.lab.uq.edu.au/pkcs/> accessed on 21 May 2026) and SwissADME online tools (<https://www.swissadme.ch/> accessed on 21 May 2026) (Pires et al. 2015, Daina et al. 2017, Hassan et al. 2024).

2.13. Statistical analysis

Data were evaluated using SPSS software V. 20.0, with means and standard errors presented for quantitative variables. The Shapiro-Wilk test was employed to appraise the normality of the data distribution. One-way ANOVA was performed, followed by Tukey's post hoc test for

statistical comparisons of the experimental groups' means. A p-value of less than 0.05 was considered to signify statistical significance. The graphs were generated using GraphPad Prism software (V. 6.01).

3. Results

3.1. GC-MS phytochemical profiling of MOLE

The GC-MS chromatogram of the ethanolic MOLE displayed several peaks, indicating the presence of various phytochemical compounds (Fig. 1). Data about the different phytochemical compounds in MOLE, including their names, retention times, peak area percentages, molecular formulas, and molecular weights are shown in Table 2. Among these, hexadecenoic acid ethyl ester, butyl-9,12-octadecadienoate, n-propyl-9,12-octadecadienoate, and methyl-7,12-octadecadienoate are determined as predominant bioactive compounds, exhibiting area ratios of 17.16 and 11.39%, respectively, with corresponding retention times of 19.91 and 22.20, respectively.

3.2. MOLE improved neurobehavioral deficits induced by CPZ

Motor coordination and balance were assessed employing the

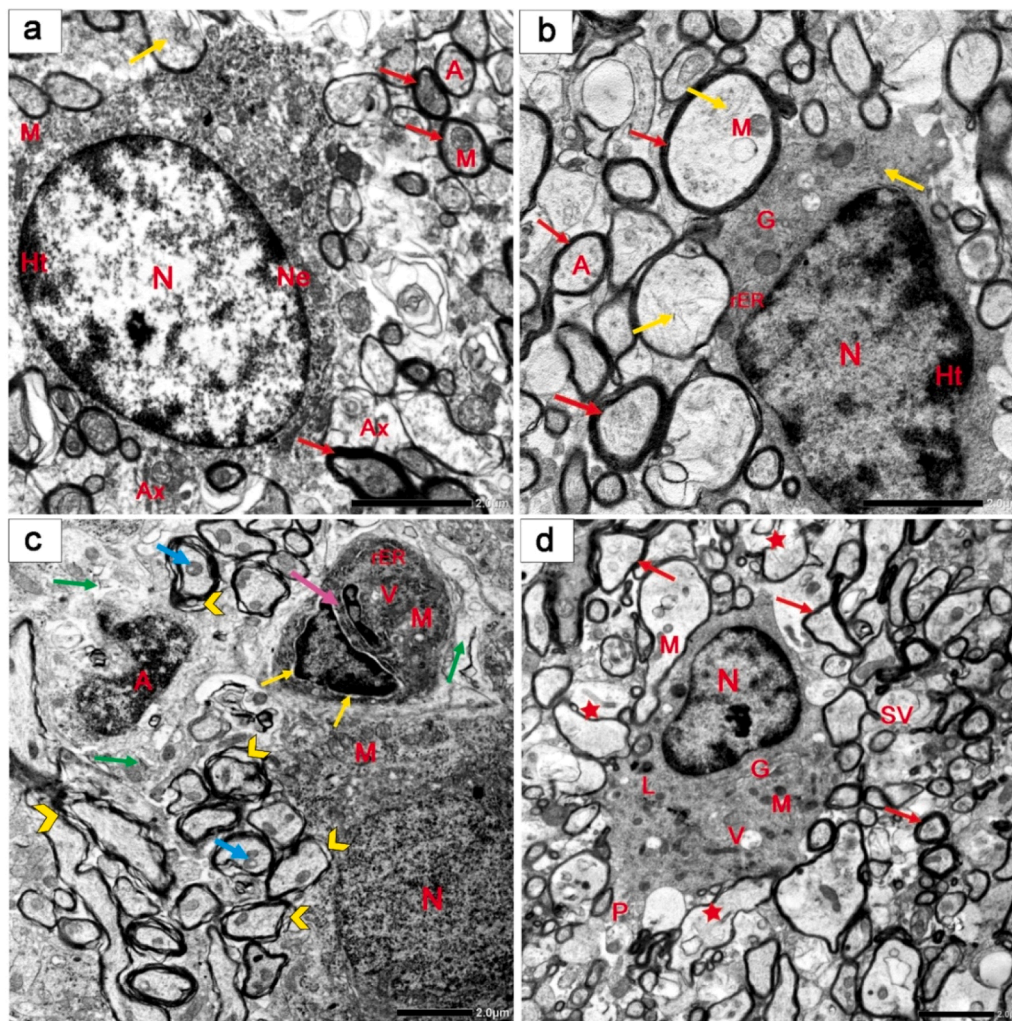


Fig. 7. Transmission electron micrographs (TEM) of the CC tissue from the brains of rats in different animal groups, showing the ultrastructural features of OLs. **Figs. (7a and 7b)** show normal OLs in the control and the MOLE groups, respectively. The OLs are depicted by their characteristic nuclei (N) associated with agglomerated heterochromatin (Ht), and a distinct nuclear envelope (Ne). The electron-dense cytoplasm encompasses mitochondria (M), rough endoplasmic reticulum (rER), and Golgi apparatus (G). Axons (A), and axon terminals (Ax) are also noticed. The myelinated nerve fibers (red arrows) contain mitochondria (M) and neurofilaments (yellow arrows). **Fig. (7c)** The micrographs of the CPZ-treated group showed a damaged OL with an enlarged perinuclear cisterna (yellow arrow), fragmented nucleus (pink arrow), mitochondria with complete loss of cristae (M), enlarged rough endoplasmic reticulum (rER), and vacuolated cytoplasm (V). Apoptotic OL (A) is also shown. Other features of CPZ intoxication include axonal demyelination (green arrows), lamellae displacement and presence of gaps in the myelin sheath (arrowheads), degenerated and vacuolated mitochondria in the nerve fibers (blue arrows), an apoptotic neuron with clear mitochondrial damage (M), and an apoptotic nucleus (N). **Fig. (7d)** CPZ plus MOLE-treated group showed restoration of normal OLs ultrastructure, illustrated by normal mitochondria (M), Golgi apparatus (G), lysosomes (L), OL process (P), and some vacuoles (V). Thinly myelinated nerve fibers (asterisks) and compact myelin wrapping the nerve fibers (red arrows) containing synaptic vesicles (SV) were also observed. Scale bars = 2 μ m.

rotarod test after three weeks of CPZ administration (**Figs. 2A and 2B**). Compared with the control group, CPZ-exposed rats revealed a marked diminution in latency to fall by 80.24% ($p < 0.001$), accompanied by a 21-fold increase in the number of falls recorded during the test ($p < 0.001$). However, when MOLE was co-administered with CPZ, the rats maintained their position on the rotarod for a significantly longer duration, reporting 212.2% increase ($p < 0.001$) in latency to fall and 83.3% reduction ($p < 0.001$) in the number of falls in relation to the CPZ-only treated group. Furthermore, muscle strength and grip performance were evaluated using the inverted screen test (**Figs. 2C and 2D**). In comparison to the control group, CPZ treatment notably diminished ($p < 0.001$) the time rats were able to remain attached to the inverted screen by 60.32%, accompanied by a 50% decrease ($p < 0.001$) in the performance score, reflecting impaired ability to remain attached to the screen and lessened muscular strength. By contrast, co-administration of MOLE markedly ameliorated these deficits, as evidenced by 171.40% increase in holding time and 100% improvement in performance score

compared with the CPZ-treated group ($p < 0.001$). Administration of MOLE alone revealed no behavioral alterations compared to the control group.

3.3. MOLE ameliorated CPZ-induced disorders in CC oxidative and antioxidative status

The oxidative and antioxidative status of the CC under the influence of CPZ and MOLE is portrayed in **Fig. 3**. CPZ treatment elicited a notable increase ($p < 0.001$) in MDA and NO levels by 218.7% and 272%, respectively, in relation to the control animals (**Figs. 3A and 3B**). This CPZ-induced deregulation of oxidative status was significantly counteracted by MOLE co-administration, resulting in reductions of MDA and NO levels by 50.7% and 56.6%, respectively, compared with animals treated with CPZ alone ($p < 0.001$). Furthermore, CPZ administration gave rise to a significant decline in the antioxidative capacity of the CC tissue (**Fig. 3C-E**). Specifically, compared to control rats, CPZ-treated

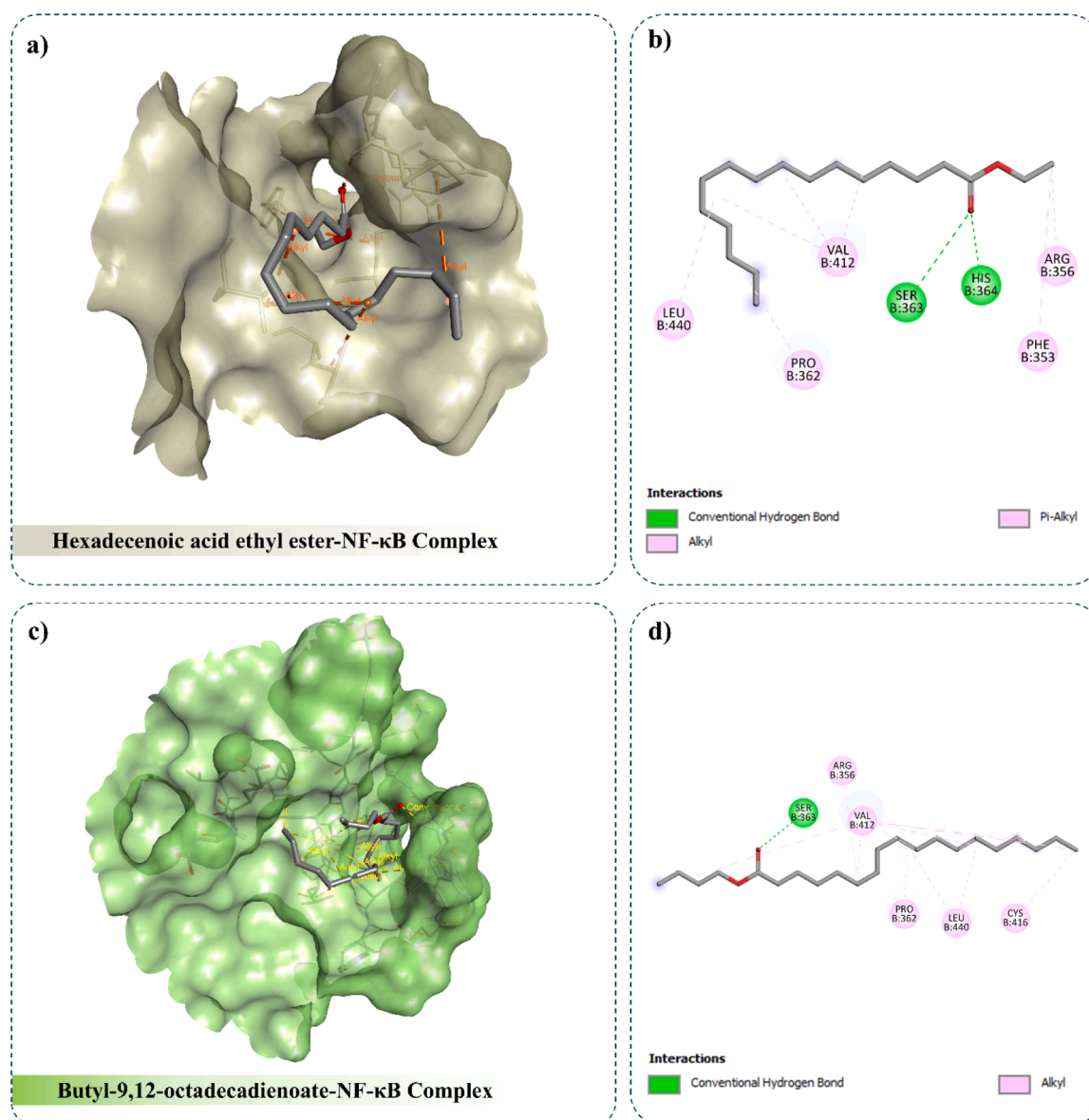


Fig. 8. Docking analysis of NF-κB. (a, b) 3D visualization and 2D interaction diagrams of hexadecenoic acid ethyl ester. (c, d) 3D visualization and 2D interactions of butyl-9,12-octadecadienoate.

rats exhibited a notable decrease in GSH level ($p < 0.001$) by 59.34%, along with reduced SOD and CAT activities ($p < 0.001$) by 43.62% and 84%, respectively. Conversely, these alterations were markedly attenuated by co-treatment with MOLE, which markedly enhanced the antioxidant status of the CC tissues, reporting increases of 68.5%, 34.7%, and 228.1% in GSH, SOD, and CAT, respectively, compared to the rats treated with CPZ alone ($p < 0.001$). Interestingly, Rats receiving MOLE alone did not show any significant differences in oxidative/antioxidant parameters relative to the control rats, suggesting a favorable safety profile for the plant extract.

3.4. MOLE improved CPZ-induced dysregulation in pro-inflammatory cytokines

To assess the anti-inflammatory potential of MOLE, the levels of inflammatory markers were quantified in CC tissue. Considering CPZ-treated rats, anti-inflammatory quantifications exhibited substantial rises ($p < 0.001$) in TNF- α , IL-1 β , IL-6, and NF- κ B by 123.4%, 20.78%, 213.3%, and 348.9%, respectively, compared to the control group as

illustrated in Fig. 4. On the contrary, co-treatment with MOLE significantly attenuated these increases, leading to reductions of 39.9%, 23.5%, 44.60%, and 37.5% in TNF- α , IL-1 β , IL-6, and NF- κ B levels, respectively, in comparison with group treated only with CPZ ($p < 0.001$), suggesting the efficacy of MOLE in mitigating CPZ-instigated inflammatory responses. Interestingly, MOLE co-treatment restored the TNF- α level in CPZ-exposed rats to a level close to that of control rats, with no statistically significant difference observed. Remarkably, no significant differences in TNF- α , IL-1 β , IL-6, or NF- κ B levels were detected between the MOLE-only and control groups, further supporting the safety profile of the extract. Collectively, our findings substantiate the potency of MOLE in abating CPZ-induced neuro-inflammation in CC tissue.

3.5. MOLE improved CPZ-induced histological aberrations in the CC tissue

Microscopic examination of toluidine blue-stained semithin sections of CC tissues demonstrated that both the control and MOLE-treated

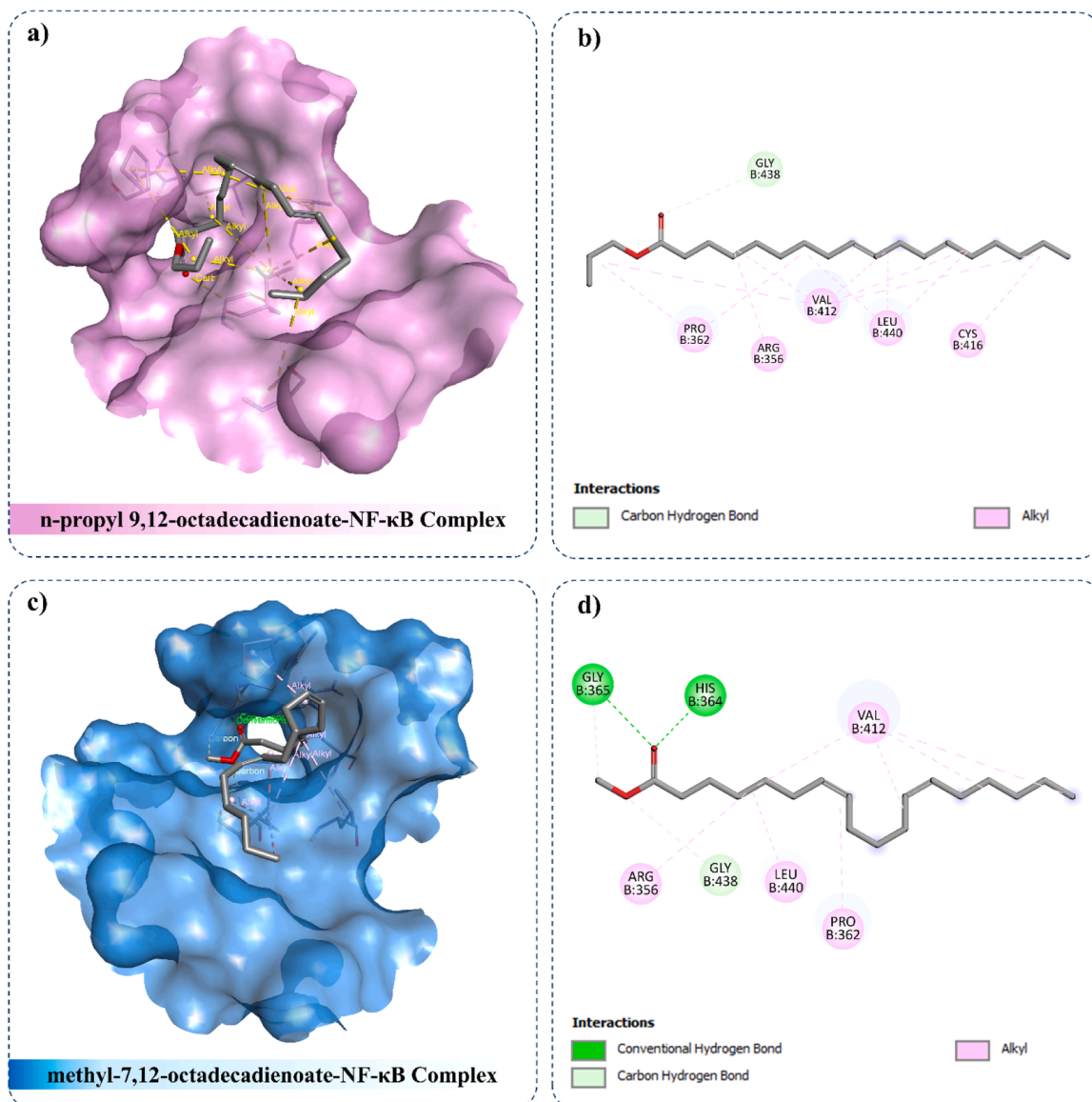


Fig. 9. Docking analysis of ligands with NF-κB. (a, b) 3D representation and 2D interaction profiles of n-propyl-9,12-octadecadienoate. (c, d) 3D and 2D interaction diagram of methyl-7,12-octadecadienoate.

groups exhibited typical histological architecture, with densely distributed myelinated nerve fibers, thick compact myelin sheaths, and abundant morphologically intact glial cells (Figs. 5a and 5b). In contrast, the CPZ-treated group showed marked pathological manifestations linked to myelin degeneration, extensive demyelinated nerve fibers, numerous pyknotic glial cells, gaps in the vicinity of glial cells, and degenerated axons displaying abnormal shapes (Fig. 5c). Notably, co-treatment with CPZ and MOLE markedly mitigated these histopathological anomalies, as evidenced by remyelination of most nerve fibers and restoration of normal glial cell morphology in the CC region as portrayed in Fig. 5d.

3.6. Restoration of CPZ-induced ultrastructural malformations in the CC by MOLE treatment

To provide further insights into MOLE-induced axonal remyelination, TEM micrographs of CC sections from different animal groups were inspected. The CC tissue from both control and MOLE-treated rats displayed myelinated nerve fibers. Specifically, normal ultrastructural characteristics of the nerve fibers were observed with darkly stained,

compact, and parallel myelin lamellae, axoplasm containing neurofilaments, and normal mitochondria (Figs. 6a and 6b). Additionally, OLs exhibited normal ultrastructural elements, including characteristic nuclei with clumped heterochromatin and a well-defined nuclear envelope. Furthermore, their electron-dense cytoplasm contained intact mitochondria, rough endoplasmic reticulum, and Golgi apparatus. In addition, axons and axon terminals were discerned. The myelinated nerve fibers illustrated with red arrows contain mitochondria and neurofilaments (Figs. 7a and 7b). In contrast, several pathological changes emerged in the CPZ-treated group (Fig. 6c), characterized by complete and partial axonal demyelination, lamellae displacement marked by the presence of gaps, and axonal damage. Moreover, the mitochondria appeared vacuolated with indistinct cristae. It is also apparent from Fig. 7c that the CPZ-treated group showed impaired OL with an enlarged perinuclear cisterna, fragmented nucleus, mitochondria with complete loss of cristae, enlarged rough endoplasmic reticulum, and vacuolated cytoplasm. Additionally, an apoptotic OL is also shown. Moreover, other features of CPZ intoxication, including axonal demyelination, lamellae displacement, gaps in the myelin sheath, degenerated and vacuolated mitochondria in nerve fibers, an apoptotic neuron with clear

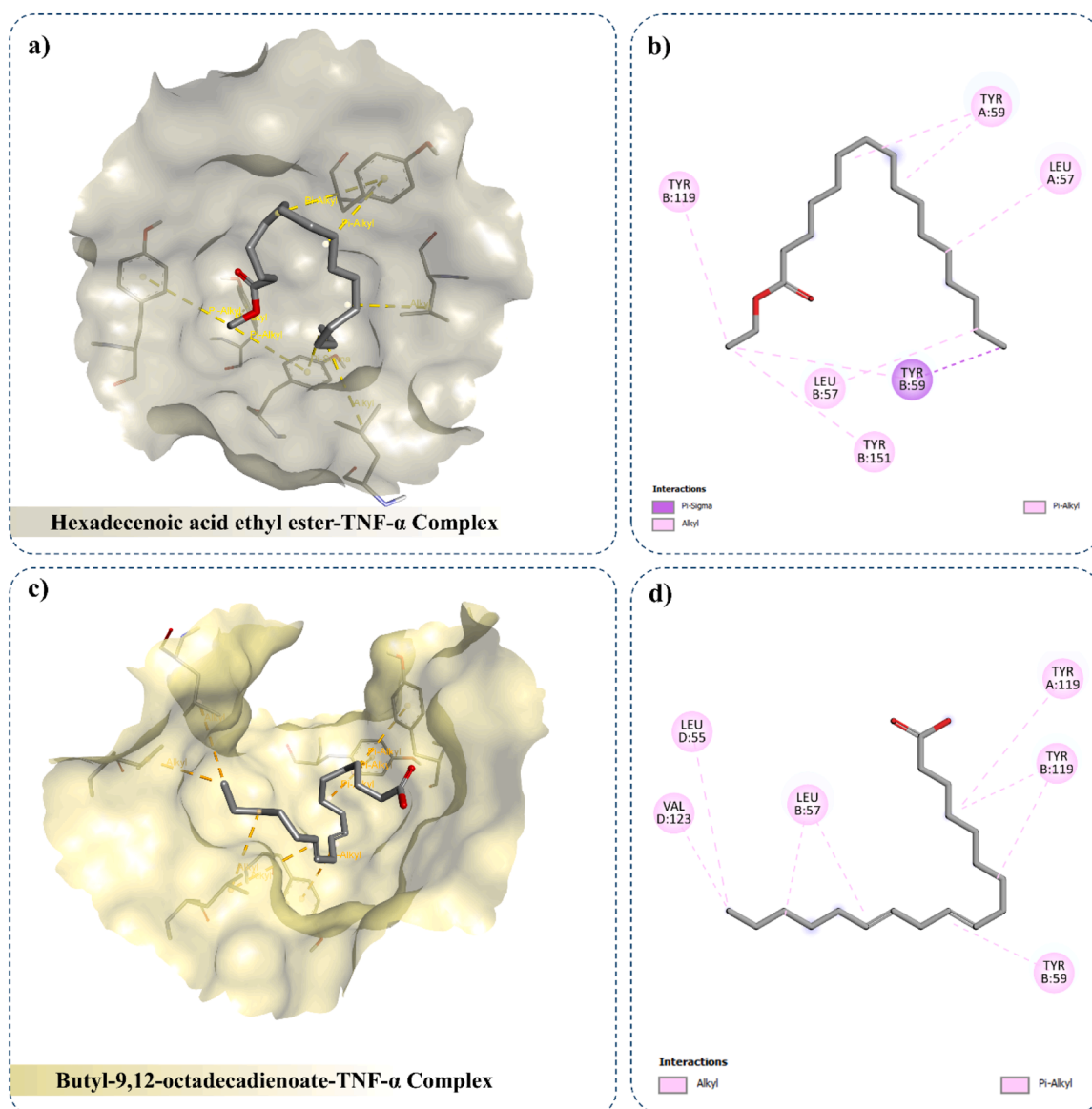


Fig. 10. Docking analysis of TNF- α . (a, b) 3D visualization and 2D interaction diagrams of hexadecenoic acid ethyl ester. (c, d) 3D visualization and 2D interactions of butyl-9,12-octadecadienoate.

mitochondrial damage, and an apoptotic nucleus were perceived. Conversely, MOLE treatment plus CPZ led to remyelination of nerve fibers, as evidenced by thinly myelinated fibers with tight and compact myelin, synaptic vesicles, and normal mitochondria. A part of an OLS' nucleus is also detected (Fig. 6d). Furthermore, restoration of normal OLS' ultrastructure, illustrated by normal mitochondria, Golgi apparatus, lysosomes, OL process, and some vacuoles was discerned. Moreover, thinly myelinated nerve fibers and compact myelin wrapping the nerve fibers containing synaptic vesicles were also observed (Fig. 7d).

Notably, Fig. 6e-g demonstrate that the CPZ-treated group revealed a substantial reduction of 64.43% ($p < 0.0001$) in myelin sheath thickness and a remarkable increase in the G-ratio by 22.22% ($p < 0.0001$) compared to the control group, whereas the axonal diameter showed no significant difference among the experimental groups. By contrast, co-treatment with MOLE and CPZ markedly augmented myelin sheath thickness by 126.5% ($p < 0.0001$) and remarkably lowered the G-ratio by 14.24% ($p = 0.0018$) compared with the CPZ-treated group.

3.7. Molecular docking analysis

Molecular docking analysis revealed that the major bioactive compounds of MOLE interact with key residues within the DNA-binding interface of NF- κ B p50/p65, suggesting potential interference with NF- κ B transcriptional activity. Hexadecenoic acid ethyl ester formed two hydrogen bonds with SER363 and HIS364, and seven hydrophobic interactions (Alkyl and Pi-Alkyl) with PRO362, VAL412, LEU440, ARG356, and PHE353 residues as displayed in Figs. 8a and 8b. Similarly, butyl-9,12-octadecadienoate formed one hydrogen bond with SER363 and eleven hydrophobic interactions (Alkyl) with ARG356, PRO362, VAL412, LEU440, VAL412, and CYS416 (Figs. 8c and 8d).

While n-Propyl-9,12-octadecadienoate established one hydrogen bond with GLY438 and twelve hydrophobic interactions (Alkyl) with ARG356, PRO362, VAL412, CYS416, and LEU440 as depicted in Figs. 9a and 9b. Methyl-7,12-octadecadienoate formed four hydrogen bonds with HIS364, GLY365, and GLY438 and seven hydrophobic interactions (Alkyl) with ARG356, PRO362, VAL412, and LEU440 (Figs. 9c and 9d).

Docking simulations also indicated that the compounds interact with the TNF- α active site residues. Hexadecenoic acid ethyl ester exhibited

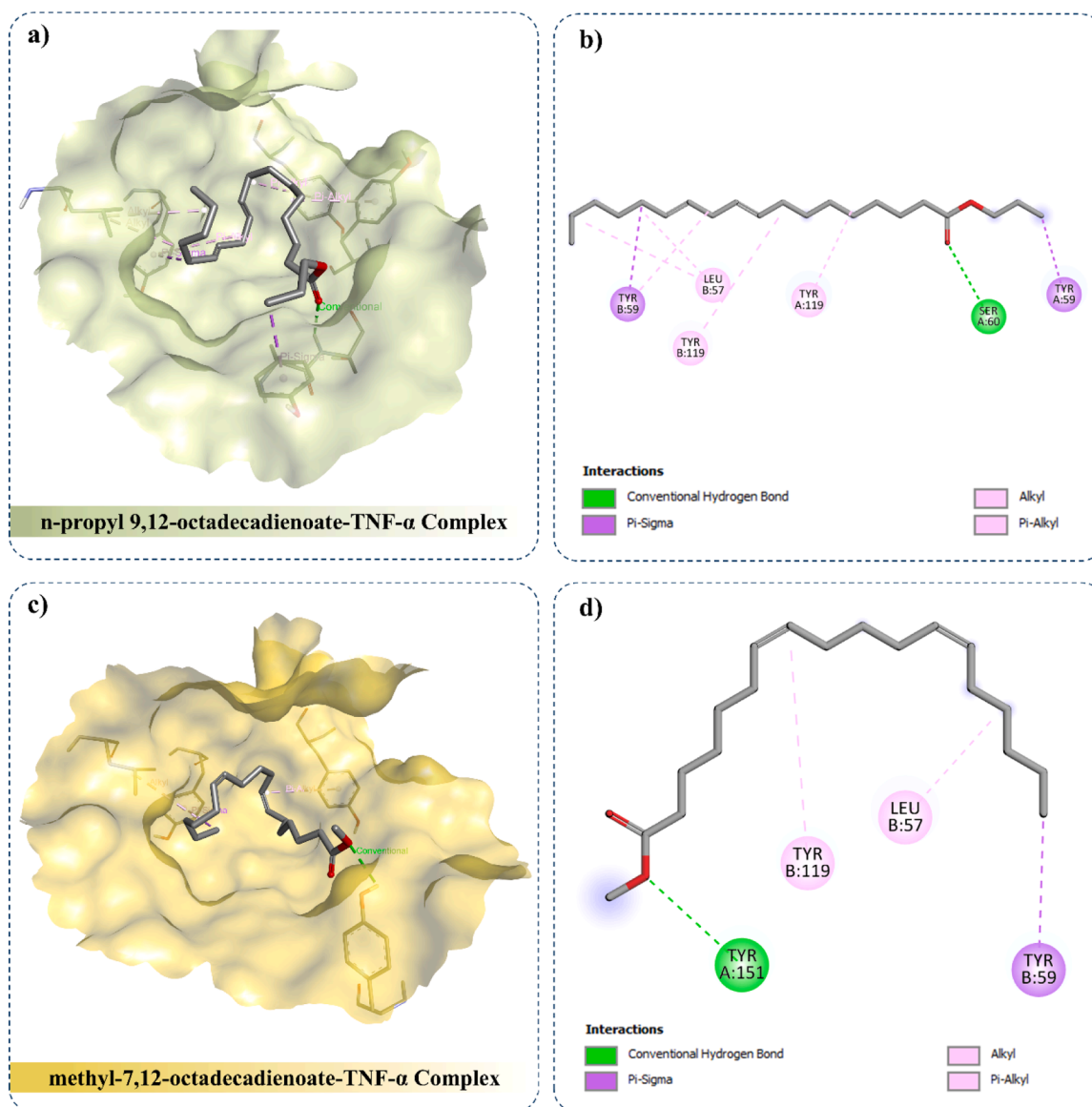


Fig. 11. Docking analysis of ligands with TNF- α . (a, b) 3D representation and 2D interaction profiles of n-propyl-9,12-octadecadienoate. (c, d) 3D and 2D interaction diagram of methyl-7,12-octadecadienoate.

eight hydrophobic interplays (Pi-Sigma, Alkyl, and Pi-Alkyl) involving TYR59, TYR119, TYR151, and LEU57 residues as illustrated in Figs. 10a and 10b. Similarly, butyl-9,12-octadecadienoate formed eight hydrophobic interactions (alkyl and Pi-Alkyl) with TYR59, TYR119, LEU55, LEU57, and VAL123 (Figs. 10c and 10d).

While n-Propyl-9,12-octadecadienoate established one hydrogen bond with SER60 and seven hydrophobic interactions (Pi-Sigma, Alkyl, and Pi-Alkyl) with TYR59, TYR119, and LEU57 as depicted in Figs. 11a and 11b. Methyl-7,12-octadecadienoate formed one hydrogen bond with TYR151 and three hydrophobic interactions (Pi-Sigma, Alkyl, and Pi-Alkyl) with TYR59, TYR119, and LEU57 (Figs. 11c and 11d). The binding affinities and detailed interaction profiles of the four compounds docked with NF- κ B and TNF- α are summarized in Table 3.

3.8. *In silico* pharmacokinetic investigation analysis

In silico ADMET and drug-likeness analyses revealed that all four bioactive compounds exhibited favorable and largely comparable pharmacokinetic properties, as summarized in Table 4. Precisely, all compounds demonstrated high absorption potential, characterized by

Caco-2 permeability and intestinal absorption values exceeding 91%, with none predicted to act as substrates for P-glycoprotein (P-gp) efflux. Distribution parameters indicated their capacity to penetrate the blood-brain barrier (BBB) and the central nervous system (CNS). Considering metabolism, all four compounds could be classified as CYP3A4 substrates and showed no inhibitory activity toward CYP2C19, CYP2C9, CYP2D6, or CYP3A4. However, all compounds exhibited potential inhibitory capacity against CYP1A2. Excretion profiles showed total clearance values ranging from 1.912 to 2.128 log mL/min/kg, with no active transport mediated by renal organic cation transporter 2 (OCT2). Toxicity evaluation showed that all molecules were non-mutagenic according to the AMES test and exhibited no hepatotoxicity. Notably, butyl-9,12-octadecadienoate was the only compound predicted to act as a hERG II inhibitor. Most importantly, all four bioactive compounds demonstrated adherence to Lipinski's Rule of Five, suggesting their favorable drug-likeness characteristics.

4. Discussion

Demyelinating disorders are complex neurological conditions

Table 3

Summary of binding affinities and molecular interactions of the selected ligands with NF- κ B and TNF- α .

Receptor	Ligand	Binding Affinity (kcal/mol)	Interacting Residue	Interaction Type
NF- κ B	hexadecenoic acid ethyl ester	-4.4	^a B:SER363:HN B:HIS364:HN B:PRO362, B: VAL412, B: LEU440, B: ARG356, B: PHE353	H-Bond H-Bond Alkyl Pi-Alkyl
	butyl-9,12-octadecadienoate	-5.2	B:SER363:HN B:ARG356, B: PRO362, B: VAL412, B: LEU440, B: VAL412, B: CYS416	H-Bond Alkyl
	n-propyl-9,12-octadecadienoate	-5.1	B:GLY438:CA B:ARG356, B: PRO362, B: VAL412, B: CYS416, B: LEU440	Carbon H-bond Alkyl
	methyl-7,12-octadecadienoate	-4.9	B:HIS364:HN, B: GLY365:HN B:GLY438:CA, B: GLY365:O B:ARG356, B: PRO362, B: VAL412, B: LEU440	H-bond Carbon H-bond Alkyl
	hexadecenoic acid ethyl ester	-5.0	B:TYR59 ^a A:LEU57, B: LEU57 A:TYR59, B: TYR59, B: TYR119, B: TYR151	Pi-Sigma Alkyl Pi-Alkyl
TNF- α	butyl-9,12-octadecadienoate	-5.4	B:LEU57, ^a D: LEU55, D: VAL123 A:TYR119, B: TYR59, B: TYR119	Alkyl Pi-Alkyl
	n-propyl-9,12-octadecadienoate	-5.4	A:SER60 A:TYR59, B: TYR59 B:LEU57 A:TYR119, B: TYR59, B: TYR119	H-Bond Pi-Sigma Alkyl Pi-Alkyl
	methyl-7,12-octadecadienoate	-5.5	A:TYR151 B:TYR59 B:LEU57 B:TYR119	H-Bond Pi-Sigma Alkyl Pi-Alkyl

^a Note: A = Chain A, B = Chain B, D = Chain D.

characterized by myelin loss, oligodendrocyte injury, glial activation, impaired remyelination, axonal damage, and disruption of blood-brain barrier (BBB) integrity (Lassmann, 2018). These deregulations are mediated by immune and inflammatory responses, which detrimentally affect the formation and maintenance of myelin sheath integrity. Herein, we investigated the therapeutic efficacy of MOLE prepared from *M. oleifera* leaves collected in Egypt on CPZ-induced demyelination in rats. Notably, CPZ administration triggered a demyelinating-like phenotype in rats, as reflected by motor impairment, oxidative imbalance, neuroinflammation, and noticeable histological and ultrastructural aberrations in the corpus callosum (CC). Administering MOLE for 3 weeks notably alleviated these disturbances, suggesting that its protective role is mainly modulated by antioxidant and anti-inflammatory

mechanisms. Furthermore, molecular docking results supported the potential interactions of the major bioactive compounds derived from MOLE with pro-inflammatory targets, including NF- κ B and TNF- α active-site residues, via multiple interplays.

The phytochemical profile of *M. oleifera* may account for its perceivable neuroprotective attributes. Over 330 phytochemicals were identified in various parts of the plant, including alkaloids, flavonoids, anthraquinones, vitamins, glycosides, sterols, esters, fatty acids, polyunsaturated fatty acids, and terpenes, although their abundance varies with environmental conditions, extraction methods, and plant part (Villegas-Vazquez et al. 2025). In the current study, GC-MS analysis of MOLE revealed different bioactive constituents, with hexadecenoic acid ethyl ester, butyl-9,12-octadecadienoate, *n*-propyl-9,12-octadecadienoate, and methyl-7,12-octadecadienoate being the most abundant. Nair et al. (Nair D et al. 2021) demonstrated that phytochemicals derived from *M. oleifera* extracts can traverse the BBB, emphasizing the biological plausibility of a direct CNS effect. Thus, the observed neuroprotective activity of MOLE in this investigation is potentially correlated with the synergistic influence of these compounds, stemming from their antioxidant and anti-inflammatory properties.

Our behavioral findings in rats following *in vivo* investigations are consistent with the structural vulnerability of the CC in CPZ-induced demyelination. Previous reports exhibited that demyelination in CPZ models is commonly accompanied by behavioral deficits and that CC impairment can undermine motor coordination (Berlucchi, 2012). In agreement with this observation, behavioral assessments of CPZ-treated rats in this study revealed impaired motor coordination and reduced limb-gripping strength, as evidenced by decreased latency times in the rotarod and inverted screen tests. Correspondingly, a similar behavioral deficiency was previously observed, particularly diminished motor coordination, following CPZ administration (Ghaiad et al. 2017, Essawy et al. 2025), although other investigations reported only a slight abnormality in rotarod performance (Xiu et al. 2017), which is likely due to variations in the extent of demyelination despite similar CPZ dosages. Concurrent administration of MOLE ameliorated behavioral deficits, as evidenced by improved motor coordination, increased muscle strength, and gripping ability, suggesting maintenance of neuronal function. This improvement may be attributed to the intrinsic antioxidant and anti-inflammatory attributes of MOLE, which may contribute to the preservation of cellular homeostasis and the promotion of remyelination in CC.

Biochemically, oxidative stress resulting from ROS overflow is recognized as a dominant mechanism underlying CPZ-induced aberrations in biochemical and structural properties in this model. Specifically, it is also acknowledged that OLs and neurons are particularly prone to oxidative injury due to their high metabolic activity and relatively insufficient antioxidant defenses. Our findings demonstrated a significant rise in levels of lipid peroxidation and NO, with a notable decrease in GSH level and SOD and CAT performances in the CC of rats treated with CPZ, implying dysregulation in redox homeostasis. These results are consistent with earlier observations (Ghaiad et al. 2017, Ibrahim Fouad et al. 2023), substantiating the implication of oxidative stress in demyelination-related pathogenesis. Additionally, a prior study reported that CPZ elicits copper deficiency, which may hinder the activity of copper-dependent enzymes, such as SOD, thereby engendering mitochondrial dysfunction, disrupting the antioxidant system, increasing ROS production in OLs, and ultimately provoking apoptosis (Nicola et al. 2024). Moreover, lipid peroxidation may particularly contribute to CPZ-induced demyelination, given that the myelin sheath is highly enriched in lipids, accounting for approximately 70%–85% of its dry mass (Kawade et al. 2026). Overall, this oxidative stress mars astrocyte and microglial function, engendering oxidative impairment of lipids, proteins, and nucleic acids as a distinct characteristic of actively demyelinated lesions, particularly around reactive astrocytes (Kim et al. 2026). Interestingly, MOLE's capacity to reduce MDA and NO levels while restoring antioxidant defenses suggests that it effectively

Table 4

Predicted Pharmacokinetic properties and drug-likeness of MOLE major bioactive compounds.

Test	Hexadecenoic acid ethyl ester	Butyl-9,12-octadecadienoate	n-propyl-9,12-octadecadienoate	Methyl-7,12-octadecadienoate	Unit
Caco-2 Permeability	1.596	1.13	1.419	1.612	log Papp in 10–6 cm/s
Intestinal absorption	91.916	91.569	92.135	92.66	% Absorbed
P-gp substrate	No	No	No	No	Yes/No
BBB permeability	0.759	0.81	0.794	0.767	log BB
CNS permeability	-1.777	-1.458	-1.515	-1.463	log PS
CYP2D6 substrate	No	No	No	No	Yes/No
CYP3A4 substrate	Yes	Yes	Yes	Yes	Yes/No
CYP1A2 inhibitor	Yes	Yes	Yes	Yes	Yes/No
CYP2C19 inhibitor	No	No	No	No	Yes/No
CYP2C9 inhibitor	No	No	No	No	Yes/No
CYP2D6 inhibitor	No	No	No	No	Yes/No
CYP3A4 inhibitor	No	No	No	No	Yes/No
Total Clearance	1.912	2.128	2.092	2.031	log mL/min/kg
Renal OCT2 substrate	No	No	No	No	Yes/No
AMES toxicity	No	No	No	No	Yes/No
Hepatotoxicity	No	No	No	No	Yes/No
Oral Rat Acute Toxicity (LD50)	1.664	1.506	1.588	1.617	mol/kg
Oral Rat Chronic Toxicity (LOAEL)	3.019	3.165	3.107	3.004	log mg/kg bw/day
hERG I inhibitor	No	No	No	No	Yes/No
hERG II inhibitor	No	Yes	No	No	Yes/No
Lipinski's Rule	Yes	Yes	Yes	Yes	Yes/No

counterbalances the repercussions of oxidative impairment and preserves neuronal and glial integrity. This antioxidant effect may be linked to its enrichment in bioactive compounds. In support of this interpretation, Moringa ethanolic extract has previously been shown to boost neuronal growth, early differentiation, dendritic and axonal development, thereby promoting neurogenesis and synaptogenesis (Hannan et al. 2014).

Neuroinflammation is another prominent feature of CPZ-induced demyelination in the present study. Thus, CPZ treatment raised TNF- α , IL-1 β , IL-6, and NF- κ B concentrations in the CC, corroborating activation of a strong inflammatory response, in agreement with previous findings (Khalilian et al. 2021, Zhang et al. 2025). Impaired BBB integrity can exacerbate demyelinating disorders by facilitating immune cell infiltration into the CNS, which amplifies local inflammation (Boutitah-Benyaich et al. 2025). In this context, microglial activation is associated with increased expression of major histocompatibility complex (MHC) class I and II molecules, induction of cell-surface co-stimulatory molecules, and release of cytokines, all of which contribute to the invasion of T lymphocytes, macrophages, and dendritic-like cells into sites of injury (Dargahi et al. 2017, Boutitah-Benyaich et al. 2025). These infiltrating cells further release pro-inflammatory cytokines, nitric oxide, and matrix metalloproteinases, aggravating BBB dysfunction and promoting continued myelin injury and neuronal impairment (Dargahi et al. 2017, Boutitah-Benyaich et al. 2025). In this inflammatory context, NF- κ B is a central transcription factor that orchestrates the production of numerous inflammatory mediators and serves as a key driver of demyelination progression (Hoffmann et al. 2025, Naserpour et al. 2025). Notably, the reduction in cytokine and NF- κ B levels after MOLE treatment emphasizes a substantial anti-inflammatory effect that likely contributes to CC protection. The depletion of NF- κ B levels in the CC may be elucidated by MOLE's potency to thwart NF- κ B nuclear translocation from the cytoplasm, thereby diminishing the release of pro-inflammatory cytokines (Mundkar et al. 2022).

Microscopic observations, encompassing histological and ultrastructural features of CC tissue, substantiate the neuroprotective role of MOLE in mitigating CPZ-induced demyelination. In accordance with previous studies (Nicola et al. 2024, Essawy et al. 2025), CPZ provoked marked axonal degeneration and myelin disruption in the CC tissues. These demyelinating effects may be largely related to OLS' impairment triggered by altered intracellular copper homeostasis, inflammatory signaling, mitochondrial dysfunction, and ROS accumulation, which is

consistent with biochemical findings (Nair D et al. 2021, Nicola et al. 2024). In addition to the oxidative and inflammatory damage described above, CPZ may impair copper-dependent cytochrome oxidase activity and reduce oxidative phosphorylation, thereby accelerating oligodendrocyte degeneration and demyelination (Kalman et al. 2007). Accordingly, mitochondrial anomalies and axonal degeneration were observed in the CC tissues of CPZ-administered rats, suggesting mitochondrial dysfunction as a potential underlying mechanism of demyelination, given its association with energy deficits, ROS overflow, and disruption of various biomolecules (Nair D et al. 2021, Nicola et al. 2024). By contrast, MOLE sustained the regular architecture of the CC and promoted remyelination, indicating preservation of OLS' structure and function. Similar myelin-enhancing effects of *M. oleifera* were reported in the CC and cerebral cortex of vanadium-treated mice (Igado et al. 2020), as well as in the hippocampus and prefrontal cortex of CPZ-administered rats (Gabriel Olaiya et al. 2018). These observations reinforce our exposition that the neuroprotective effects of MOLE are linked to the suppression of oxidative stress and inflammation, along with enhanced cellular conditions that boost myelin preservation and repair (Hannan et al. 2014).

Most importantly, the remarkable augmentation in the G-ratio accompanied by a reduction in myelin thickness without alterations in axon diameter stemmed from myelin loss or thinning rather than alterations in axonal caliber due to axonal shrinking or swelling. These findings suggest that CPZ-induced pathology predominantly targets the myelin sheath while maintaining axonal structure (Baxi et al. 2015). These ultrastructural findings are consistent with selective demyelination provoked by OLS loss, a characteristic hallmark of the CPZ model (Zirngibl et al. 2022, Kipp, 2024). In contrast, the augmentation in myelin sheath thickness as a result of MOLE treatment emphasizes the therapeutic efficacy of the extract in enhancing remyelination. These findings are in line with a previous report that demonstrated a remarkable increase in the G-ratio in CPZ-treated mice, associated with a significant reduction in myelinated fibers compared to the control group (Nicola et al. 2024). Moreover, they reported the remyelination impact of rutin in the CPZ-treated mice, as evidenced by a pronounced reduction in the G-ratio and a substantial rise in the percentage of myelinated fibers (Nicola et al. 2024).

Alongside our experimental findings, molecular docking results reported in this work further underscore the counteractive influence of MOLE, revealing distinct biochemical interactions. In this context, the

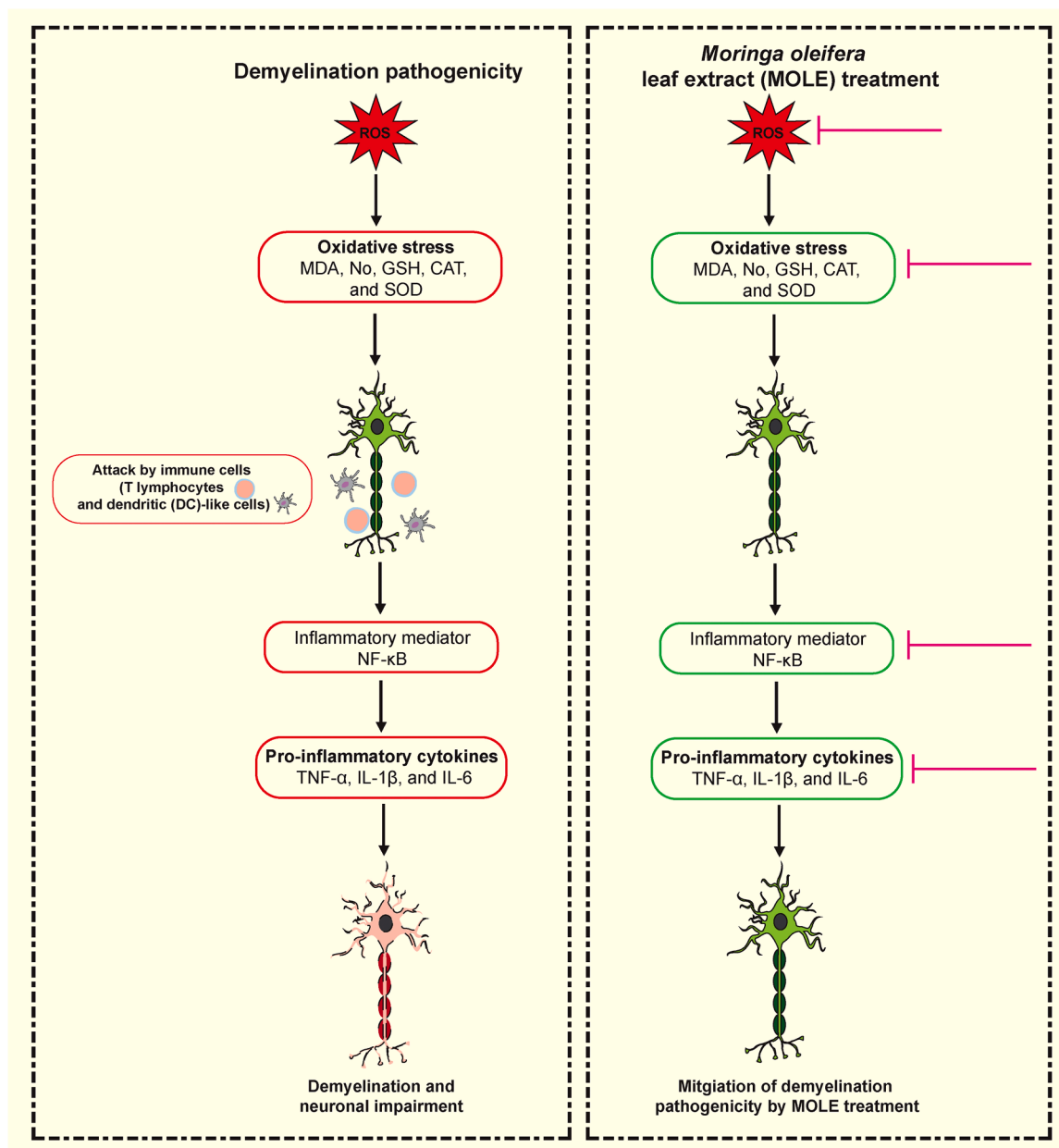


Fig. 12. Schematic demonstration depicting the potential mechanism by which MOLE counteracts CPZ-induced demyelination in corpus callosum (CC) tissues.

four most abundant bioactive compounds of MOLE, hexadecenoic acid ethyl ester, butyl-9,12-octadecadienoate, n-propyl-9,12-octadecadienoate, and methyl-7,12-octadecadienoate, showed notable bindings to the DNA-binding interface of NF-κB and the active site amino acids of TNF-α. Given that these compounds are long-chain hydrocarbon esters with predominantly nonpolar structures, they displayed broadly similar hydrophobic interaction profiles with both targets. Importantly, suppression of NF-κB and TNF-α has previously been associated with attenuation of CPZ-induced demyelination and neurodegeneration (Abdi et al. 2021, Zhao et al. 2025). Thus, the docking analysis complements the *in vivo* data, suggesting that MOLE phytochemicals may be associated with the modulation of inflammatory signaling in the CC tissue, thereby alleviating demyelination and neurodegeneration.

Furthermore, the *in silico* pharmacokinetic analysis supports the potential of these compounds as CNS-targeted neuroprotective candidates against CPZ-induced demyelination. Remarkably, their predicted pharmacokinetic profiles indicated favorable intestinal absorption and efficient BBB penetration, while suggesting a low potential for P-

glycoprotein-mediated efflux. Moreover, the favorable systemic safety profiles of these compounds, supported by the absence of AMES mutagenicity and hepatotoxicity, together with their compliance with Lipinski's rule of five, further support the *in vivo* observations in our study. Taken together, our findings suggest that MOLE mitigates CPZ-induced reversible demyelination in the CC through coordinated antioxidant and anti-inflammatory mechanisms as delineated in Fig. 12, enhancing functional and structural properties.

Despite the important and promising findings presented in this study concerning the neuroprotective role of MOLE against CPZ-induced demyelination, key limitations of this study should be acknowledged. The study lacked quantitative real-time PCR (qPCR), along with western blot and immunohistochemical analyses using specific antibodies, which could provide more compelling evidence for the therapeutic efficacy of MOLE in terms of investigating demyelination in the CPZ-treated group and remyelination in the animal group co-treated with CPZ and MOLE, as well as comprehending the molecular signaling pathways underlying its neuroprotective effects. Furthermore, further investigations at the

molecular and protein levels are warranted, including integrated proteomic, transcriptomic, and metabolomic approaches to decipher the precise mechanisms responsible for the observed neuroprotective outcomes. Such investigations may also open new avenues for identifying key target genes and proteins involved in MOLE therapeutic performance, thereby providing a robust mechanistic underpinning for its potential application in demyelinating deregulations.

5. Conclusion

To sum up, given that oxidative stress and neuroinflammation play critical roles in the pathophysiology of demyelinating disorders, this study examined the protective effects of ethanolic *Moringa oleifera* leaf extract (MOLE) against demyelination incited by CPZ in rats. GC-MS analysis of MOLE exhibited four compounds, including hexadecenoic acid ethyl ester, butyl-9,12-octadecadienoate, n-propyl-9,12-octadecadienoate, and methyl-7,12-octadecadienoate, which were identified as predominant bioactive compounds. The *in vivo* investigations demonstrated that CPZ led to a notable impairment in motor coordination and gripping strength in the rats' limbs. CPZ also disrupted redox status, increased neuroinflammation, and upregulated NF- κ B levels in the corpus callosum (CC) region. Histological and ultrastructural examinations revealed loss of myelinated fibers and death of OLs within the CC tissue. Administration of MOLE for three weeks effectively mitigated all these consequences, as evidenced by findings from behavioral, biochemical, molecular, histopathological, and ultrastructural studies. Furthermore, molecular docking analysis indicated that the major bioactive compounds of MOLE possess the potential to interact with the NF- κ B and TNF- α active site residues through various interactions. The mechanisms behind MOLE's protective effect against demyelination in the CC region are most likely due to its antioxidant and anti-inflammatory attributes. This advantageous effect of MOLE may also be associated with the modulation of the NF- κ B signaling pathway. Notably, *in silico* pharmacokinetic analysis manifested favorable safety profiles for the four major bioactive compounds defined in MOLE, consistent with the absence of adverse effects observed in rats treated only with MOLE. Additionally, all compounds complied with Lipinski's rule of five, suggesting their favorable drug-likeness attributes. Overall, given the natural origin and safety profile of MOLE, it may represent a promising therapeutic candidate or adjunctive intervention for demyelinating disorders.

CRedit authorship contribution statement

Amina E. Essawy: Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization. **Rania I. Fahmy:** Methodology, Data curation. **Rehab A. Goma:** Writing – original draft, Visualization, Supervision, Methodology, Formal analysis, Data curation. **Howida A. Fetouh:** Methodology, Formal analysis, Data curation. **Mohamed Helal:** Writing – review & editing, Data curation. **Ahmed Massoud:** Writing – original draft, Methodology, Formal analysis. **Mohamed A. Hassan:** Writing – review & editing, Visualization, Formal analysis. **Wessam M. Abdel-Wahab:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Data curation.

Ethics statement

The *in vivo* experimental procedures in this study followed ARRIVE guidelines (<https://arriveguidelines.org>) and the National Institutes of Health guidelines for the care and use of laboratory animals. The Institutional Animal Care and Use Committee (IACUC) at Alexandria University, a member of the International Council for Laboratory Animal Science (ICLAS), approved the investigation protocol (approval number: AU04230328103).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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